

The grammar of gender: Insights from Bantu asymmetries of AGR with conjoined subjects

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1. Introduction

In this paper I argue that asymmetries in agreement with conjoined nominals in Bantu provide important new insights into the nature and workings of grammatical gender and gender agreement. The empirical base is noun class morphosyntax in the Xhosa language, with some helpful additional information from related Shona. I follow Carstens (1991) among others in analyzing odd/even pairs of Bantu noun classes as singulars/plurals of 5+ genders as will be detailed in §2; Xhosa has 5 such pairs and Shona has 8 (plus a few that do not pluralize). The large inventories of genders in Bantu provide information beyond what is available in 2-3 gender languages.

My focus is a set of systematic contrasts among conjoined singulars and plurals.

In some genders, conjoined singulars pair with gender-matching plural agreement. This is the case in Xhosa examples (1)a,b where the class 2 subject marker accompanying class 1 conjuncts is the same as is required with a plural subject in (1)c. Three genders exhibit this pattern: those corresponding to noun classes 1/2, 7/8, and 9/10.^{1,2}

- (1) a. Um-mi nom-ongameli **ba**-ya-ncokola. [1+1=2]
1-citizen and.1-president 2SM-DISJ-chat
'The citizen and the president are chatting.'
- b. U-loliwe kunye no-matshini **ba**-ya-hamba. [1+1=2]
1a-train and and.1a -machine 2SM-DISJ-move
'The train and the machine are moving.'

¹ Examples are from my fieldwork unless otherwise noted. Empirical generalizations regarding Xhosa conjoined singulars are drawn from Taraldsen et al (2018) who cite 21 Xhosa speakers, and from follow-up field research of my own with 8 speakers, all quite consistent in judgments. Xhosa examples of conjoined plurals in §6 are from my fieldwork and provide numerical breakdowns of responses since there is variability.

² In glosses, numerals indicate noun class (number+gender); SM = subject marker; OM = object marker; ASP = aspect; DISJ = disjoint, a VP-final verb form; LOC = locative. PST = past tense; PRT = participle; F(em) = feminine; M(asc) = masculine; N(eut) = neuter; PL = plural; POSS= possessor. [X] denotes semantic feature X.

- c. Aba-mi/oo-matshini **b-a-wa.**
2-citizens/2a-machines 2SM-ASP-fall
'The citizens/machines fell.'

For conjoined singulars of two other classes including class 5 below, gender-matching plural agreement is not possible. Instead, agreement is class 2 for human-denoting conjuncts like (2)a, the same as for class 1 singulars like (1)a-c. For others such as the inanimates in (2)b the subject marker is class 8. (2)c shows that the class 6 *a*- subject marker which is ruled out in (2)a,b is required with plural subjects of this gender. Xhosa class 3 nouns also exhibit this pattern and as we will see in §7, six singular classes do so in Shona. The examples in (3) show that conjoined singulars of mismatched genders also pair with class 2 if [human], otherwise class 8.

- (2) a. I-gqwetha ne-sela ✓**ba**/X**a**-phumile. [5+5=2; #6]
5-lawyer and.5-thief 2SM/ 6SM-left.
'The lawyer and the thief left.'
- b. I-qanda ne-cepe X**a**-/✓**z**-a-wa. [5+5=8; #6]
5-egg and.5-spoon 6SM/8SM-ASP-fall
'The egg and the spoon fell.'
- c. Ama-sela X**ba**/✓**a**-phum-ile.
6-thieves 2SM/ 6SM-leave-PST.DISJ
'The thieves left.'
- (3) a. Is-anuse nen-tombi **ba**-ya-sebenza. b. U-bhaka nen-cwadi **zi**-ngaphandle.
7-medium and.9-girl 2SM-DISJ-work 1a-backpack and.9-book 8SM-be.outside
'The medium and the girl are working.' 'The backpack and the book are outside.'

The asymmetry of (1) vs (2) is not Bantu-particular – in Bosnian, Croatian and Serbian (BCS), feminine and masculine pattern like Xhosa class 1 while neuter gender behaves similarly to Xhosa classes 3 and 5 ((4) from Murphy & Puskar 2018; (5) from Arsenijević 2021).

- (4) [_{&P} Zavesa i biljka] su ukrašavale prozor. [BCS]
 curtain.F.SG and plant.F.SG are decorate.PRT.F.PL window
 'A curtain and a plant decorate the window.'
- (5) Sel-o i polj-e su poltopljen-i/*poltopljen-a. [BCS]
 village-NEUT.SG and field.NEUT.SG AUX.PL flooded-MASC.PL/ *NEUT.PL
 'The village and the field are flooded.'

Despić (2016), Nevins (2018), Tsimpli & Hulk (2013) propose that neuter in BCS is an absence of

gender inflection or spells out negative values [-fem, -masc]. The approach cannot be extended to Xhosa since there are two such genders, or to related Shona, where the genders whose conjoined singulars pattern like those of BCS neuter outnumber the others. Adamson & Anagnostopoulou (2024), building on (Arsenijević 2017, 2021), relate the BCS facts to neuter containing mostly mass nouns; Arsenijević shows that they are defective in certain ways. But we will see that classes 3/4 and 5/6 do not exhibit any corresponding unusual properties.

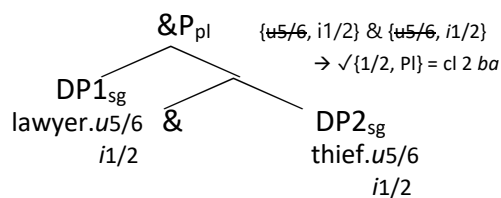
The facts of Xhosa conjoined plurals diverge from those of their singular counterparts in significant ways. When plurals match in features, agreement also matches them; (6) shows that the *ba-* subject marker is unacceptable to my consultants even for conjoined plurals like [6+6] whose singulars we saw in (2) must pair with *ba-*. When conjoined plurals mismatch in gender, first conjunct agreement (FCA) is preferred with DP1 of plural classes 2, 8, or 10 by five out of eight speakers in my study as shown in (7)a. But speakers' preference shifts to last conjunct agreement (LCA) if DP1 is of classes 4 or 6, as the data and margin notes in (7)b,c indicate.

- (6) Ama-gwetha nama-sela a-/ *ba-ya-sebenza ndawonye. [6+6=6]
6-lawers and.6-thieves 6SM/2SM DISJ-work together
'The healers and the officers are working together.'
- (7) a. lin-tombi **naba-ntwana** zi/ba-ya-cul-a. [10+2: FCA>**LCA** 5:3]
10-girls and.2-children 10SM/2SM-DISJ-sing-FV
'The young ladies and the children are singing.'
- b. Imi-gewu na-**ba-ntwana** **ba/i/zi**-ya-cul-a. [4+2: **LCA**>FCA>[-human] default 5:2:1]
4-criminals and.2-children 2SM/4SM/8SM-DISJ-sing-FV
'The criminals and the children are singing.'
- c. Ama-polisa na-**ba-ntwana** **ba/i**-ya-cul-a. [6+2: **LCA** >FCA 6:2]
6-polisa and.2-children 2SM /6SM-DISJ-sing-FV
'The policemen and the children are singing.'

My analysis adopts the *n* approach to gender and *nP* stacking from Kramer (2015; see also Fuchs & van der Wal 2022). The proposals I will advocate are the following. First, genders like BCS neuter and Xhosa 3/4 and 5/6 have no *i*(nterpretable) content that is legible in narrow

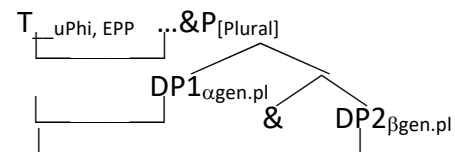
syntax; I characterize Xhosa 3/4, 5/6 as *u*(ninterpretable) genders which always stack above *icores* of 1/2 [human], 7/8 [inanimate], or 9/10 [animal]. Second, the resolution process deletes the *u*genders percolated to &P. Agreement with conjoined singulars is valued by the exposed *icores* since only *i*content is factored into resolution; in this I adapt a proposal of Adamson & Anagnostopoulou (2024) to predict the patterns of “default” agreement with Xhosa conjoined singulars as sketched in (8)a. Third, genders which have both arbitrary and non-arbitrary members are reasonably taken to have *i*- and *u*flavors (Kramer 2015), but the flavors are not distinguished in the grammar of agreement with conjuncts of Xhosa classes 1/2, 7/8, 9/10 and BCS feminine and masculine; (1)a,b and (4) are representative in this respect. I propose that a gender’s *u*- vs. *i*flavors are indistinguishable until LF licensing conditions ensure appropriate [gender+root] pairings (adapting Kramer 2015). Fourth, the facts of agreement with conjoined plurals differ systematically from those of conjoined singulars because in Xhosa, only singular conjuncts trigger resolution on &P, contra Corbett & Mtenje’s (1987) claim that gender mismatch also triggers it. For plural conjuncts, I propose an analysis involving Multiple Agree reaching into &P_{pl} to copy features of DP1 and DP2; a type of impoverishment reduces the multiple copies to one set, guided by a language-particular hierarchy of preferences which in Xhosa favor *igenders* and FCA. These proposals yield the contrasting derivations in (8)a,b.

(8) a. Agreement with conjoined singulars



Percolation; deletion of *u*genders
resolution to shared *i*features

b. Agreement with conjoined plurals



Multiple Agree (T, &P, DP1, DP2);
reduction to a single set for exponence,
favoring *igenders* and FCA

Fifth, because only 2 of Shona's 8+ genders exhibit gender-matching plural agreement with conjoined gender-matched singulars, I propose that languages with *iflavors* are generally few; languages with a small number of genders obscure this. Last but not least is my claim that the success or failure of agreement with conjoined singulars of a given gender diagnoses the existence of *iflavors* in that gender, making possible this set of insights.

This paper has nine sections. §2 presents the basics of Xhosa noun classes and agreement. §3 details the asymmetries in agreement with conjoined singulars. §4 sketches the ingredients of the analysis, and §5 applies them to derive the patterns. §6 describes and analyzes the asymmetries of conjoined plurals. §7 explores the possibility that non-canonical semantic content makes Xhosa 3/4 and 5/6 defective, applying some diagnostics from Arsenijević (2017, 2021)'s explorations of BCS neuter with negative results. §7 also shows that in Shona, only two of eight paired classes exhibit regular plural agreement with conjoined singulars and concludes that agreement failure is typical because most genders are uninterpretable and excluded in the calculus of resolved agreement. §8 concludes.

2. Xhosa basics

2.1 Noun class overview

Xhosa has five regular singular/plural pairings exemplified in (9). These pairings are generally identified as noun classes 1-10 following a numbering system that dates to Meinhof (1906) and work of Malcolm Guthrie (1948, 1971). For clarity I henceforth color-code class prefixes and the agreement that distinguishes classes in the data and/or in glosses.

- (9) a. **um**-ntu/**aba**-ntu b. **um**-pu/**imi**-pu. c. **ili**-so/**ame**-hlo d. **isi**-tya/**izi**-tya e. **in**-ja/**izin**-ja
 1-/2-person 3-/4-gun 5-/6-eye 7-/8-dish 9-/10-dog
 'person/s' 'gun/s' 'eye/s' 'dish/es' 'dog/s'

I ignore Xhosa class 11 which pluralizes in class 10 (yielding the same pattern as conjoined

singulars of class 9), and classes 14 and 15, which lack plurals; these two include mainly abstract nouns, infinitives, gerunds, and deverbal nouns.

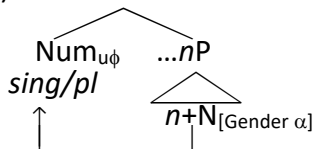
A number of researchers have argued that grammatical genders underlie each Bantu singular/plural noun class pairing (see Carstens 1991, 2008, Corbett 1991, Corbett & Mtenje 1987, Fuchs & van der Wal 2022, Watkins 1937). Example (10) shows the system of five genders A-E proposed in Carstens (1991) for noun classes 1-10. Carstens proposed that a noun enters the syntax endowed with a gender feature and raises to adjoin to the head of a Num(ber)P projection; updating this to reflect the *n* approach to gender yields (11). In Carstens's system, noun class prefixes are gender-particular spellings out of number features as shown in (12).

(10) Bantu Genders (Carstens 1991)

Gender **A**: stems of Classes 1/2
 Gender **B**: stems of Classes 3/4
 Gender **C**: stems of Classes 5/6

Gender **D**: stems of Classes 7/8
 Gender **E**: stems of Classes 9/10
 Etc.

(11) ...NumP



(12) Nouns class prefixes are gender-specific number morphology

[Singular] ↔ um- / __N _[Gender A]	[Plural] ↔ aba- / __N _[Gender A]
[Singular] ↔ isi- / __N _[Gender D]	[Plural] ↔ izi- / __N _[Gender D] etc.

I adopt Carstens's analysis of noun classes as singulars and plurals of genders A-E, but for convenience I utilize the standard numeric classes in glosses and much of the discussion.

There is abundant agreement in noun class on Xhosa nominal modifiers and predicates.

Table 1 illustrates the subject and object markers and adjectival concord for each class. I include subclasses 1a/2a as they are relevant from time to time; they have different prefixes from other nouns of classes 1/2, but agreement with them is regular 1/2 agreement as shown here.

Table 1: Xhosa noun class markers and agreement

class	noun	gloss	X _{class.α} will fall	I saw <pronoun> class.α	adjectival concord
1	um-ntu	person	u-zo-wa 1SM-FUT-fall	Nd-a-m-bona 1SSM-PST-1OM-see	um-ntu om-khulu 1-person 1-big
2	aba-ntu	people	ba-zo-wa 2SM-FUT-fall	Nd-a-ba-bona 1SSM-PST-2OM-see	aba-ntu aba-khulu 2-people 2-big
1a	u-mama	mother	u-zo-wa 1SM-FUT-fall	Nd-a-m-bona 1SSM-PST-1OM-see	u-mama om-khulu 1-mother 1-big
2a	oo-mama	mothers	ba-zo-wa 2SM-FUT-fall	Nd-a-ba-bona 1SSM-PST-2OM-see	oo-mama aba-khulu 2-mothers 2-big
3	um-thi	tree	u-zo-wa 3SM-FUT-fall	Nd-a-wu-bona 1SSM-PST-3OM-see	um-thi om-khulu 3-tree 3-big
4	imi-thi	trees	i-zo-wa 4SM-FUT-fall	Nd-a-yi-bona 1SSM-PST-4OM-see	imi-thi emi-khulu 4-trees 4-big
5	ili-tye	stone	li-zo-wa 5SM-FUT-fall	Nd-a-li-bona 1SSM-PST-5OM-see	ili-tye eli-khulu 5-stone 5-big
6	ama-tye	stones	a-zo-wa 6SM-FUT-fall	Nd-a-wa-bona 1SSM-PST-6OM-see	ama-tye ama-khulu 6-stones 6-big
7	isi-tya	dish	si-zo-wa 7SM-FUT-fall	Nd-a-si-bona 1SSM-PST-7OM-see	isi-tya esi-khulu 7-dish 7-big
8	izi-tya	dishes	zi-zo-wa 8SM-FUT-fall	Nd-a-zi-bona 1SSM-PST-8OM-see	izi-tya ezi-khulu 8-stones 8-big
9	in-komo	cow	i-zo-wa 9SM-FUT-fall	Nd-a-yi-bona 1SSM-PST-9OM-see	in-komo en-kulu 9-cow 9-big
10	iin-komo	cows	zi-zo-wa 10SM-FUT-fall	Nd-a-zi-bona 1SSM-PST-10OM-see	iin-komo ezi-khulu 10-cows 10-big

2.2 Semantics of noun classes

2.2.1 Proto-Bantu

Strands of semantic unity have led to reconstructions in which at least some Proto-Bantu classes have regular meanings associated with them. Table 2 is fairly standard in this regard.

Table 2: A Proto-Bantu noun class reconstruction (Vossen & Dimmendaal 2020:151)

Singular (number) ♦	Singular (form) ♦	Plural (number) ♦	Plural (form) ♦	Semantics ♦
1	*mũ-	2	*βà-	humans
3	*mũ-	4	*mì-	trees, plants
5	*lì-	6	*mà-	mixed/cl. 6 liquids
7	*kì-	8	*βì-	mixed
9	*nì-	10	*lì-nì-	animals, mixed
11	*lù-			mixed
12	*kà-	13	*tù-	augmentative, diminutive, etc.
14	*βù-			abstract
15	*kù-			infinitive
16	*pà-			location on
17	*kù-			location at
18	*mũ-			location in
19	*pì-			diminutive

In Xhosa noun classes there are a few semantic regularities consistent with Table 2: only [human] nouns bear the **um-/aba-** prefixes of classes 1/2, and kinship terms are consistently found in subclass 1a/2a bearing its distinct prefixes **u-/oo-**. But these are one-way correlations: [human] nouns are scattered across the classes, and in addition to kinship terms, 1a/2a include all the letters of the alphabet and an assortment of animals and inanimates. Other semantic associations reconstructed in Table 1 are weak or lost. Liquids, tree and plant terms, and animal names occur in all classes. Diminutive, augmentative, and locative classes 16 and 18 are absent in Xhosa; Nguni locative class 17 is prepositional (Buell 2007) and will not be relevant here.

The heterogeneity of Xhosa noun classes is a significant factor in the analysis. In what follows I provide an overview of the distribution of the major semantic fields across them.

2.2.2 The heterogenous contents of Xhosa classes

(13) lists some representative examples of the distribution of [human] nouns in Xhosa.

In classes 3/4 there are only a few and they are stigmatized, but in 1/2, 5/6, 7/8, and 9/10 there are very many, without predictable semantic connotations.³

(13) Humans: classes **1/2**

- | | | |
|--|--|---|
| a. um- ntwana/ aba- ntwana
1-child/2-child
'child/ren' | b. um- ntu/ aba- ntu
1-person/2-people
'person/people' | c. um-/aba- hleli
1-editor/2-editor
'editor/s' |
|--|--|---|

Humans: classes **3/4** (stigmatized)

- | | | |
|---|---|--|
| d. um- gulukudu/ imi- gulukudu
3-gangster/4-gangster
'thug/s, gangster/s' | e. um- gewu/ imi- gewu
3-criminal/4-criminal
'criminal/s' | f. um- lwelwe/ imi- lwelwe
3-/4-disabled or poor
'disabled or poor person/s' |
|---|---|--|

³ Mletshe (2019) illustrates some semantic regularities correlating with output classes of deverbal noun formation. As these are not characteristic of basic, underived nouns they are somewhat orthogonal to my concerns in this paper; space precludes full discussion but I suggest that semantic phenomena are properties of the particular nominalizers and nominalization processes involved – see Carstens (2008) for an approach.

Humans: classes 5/6

- | | | |
|--|---|--|
| g. i-qhawe/ama-qhawe
5-hero/6-hero
'hero/es' | h. i-gqwetha/ama-gqwetha
5-lawyer/6-lawyer
'lawyer/s' | i. i-sela/ame-sela
5-thief/6-thief
'thief/thieves' |
|--|---|--|

Humans: classes 7/8

- | | | |
|--|---|--|
| j. isi-bonda/izi-bonda
7-headman/8-headmen
'headman/men' | k. isi-hlobo/izi-hlobo
7-friend/8-friend
'friend/s' | l. is-anuse/iz-anuse
7-diviner/8-diviner
'diviner/s' |
|--|---|--|

Humans: classes 9/10

- | | | |
|--|---|--|
| m. in-tombi/iin-tombi
9-young.lady/10-young.lady
'young lady/ladies' | n. in-gcali/iin-gcali
9-expert/10-expert
'expert/s' | o. im-bongi/iim-bongi
9-poet/10-poet
'poet or praise singer/s' |
|--|---|--|

[Animal] nouns are also well-represented in every pair of classes, as (14) demonstrates.

Classes 9/10 seem to include a plurality of them, however.

(14) Animals: classes 1a/2a

- | | | |
|--|---|--|
| a. u-nokala/oo-nokala
1a-crab/2a-crab
'crab/s' | b. u-krebe/oo-krebe
1a-shark/2a-shark
'shark/s' | c. u-dyakalashe/oo-dyakalashe
1a-jackal/2a-jackal
'jackal/s' |
|--|---|--|

Animals: classes 3/4

- | | | |
|---|--|--|
| d. um-khombe/imi-khombe
3-rhino/4-rhino
'rhino/s' | e. um-qhagi/imi-qhagi
3-rooster/4-roosters
'rooster/s' | f. um-thwane/imi-thwane
3-donkey/4-donkey
'donkey/s' [Zulu?] |
|---|--|--|

Animals: classes 5/6

- | | | |
|---|--|---|
| g. i-cikilishe/ama-cikilishe
5-lizard/6-lizard
'lizard/s' | h. i-hashe/ama-hashe
5-horse/6-horse
'horse/s' | i. i-hobe/ama-hobe
5-dove/6-dove
'dove/s' |
|---|--|---|

Animals: classes 7/8

- | | | |
|---|--|---|
| j. isi-gcawu/izi-gcawu
7-spider/8-spider
'spider/s' | k. isi-khova/izi-khova
7-owl/8-owl
'owl/s' | l. isi-lwanyana/izi-lwanyana
7-animal/8-animal
'animal/s' |
|---|--|---|

Animals: classes 9/10 (plurality of animals; a fairly predictable mapping)

- | | | |
|--|---|--|
| m. in-ja/iin-ja
9-dog/10-dog
'dog/s' | n. in-dlovu/iin-dlovu
9-elephant/10-elephant
'elephant/s' | o. i-hagu/ii-hagu
9-pig/10-pig
'pig/s' |
|--|---|--|

[Inanimate] nouns are similarly scattered. For example, though many are found in 7/8, the generic term for 'thing(s)' is a class 9/10 noun.

(15) Inanimates: 1a/2a

a. u-lolilwe/oo-lolilwe
1a-train/2a-train
'train/s'

b. u-matshini/oo-matshini
1a-machine/2a-machine
'machine/s'

c. u-L/oo-L
1a-L/2a-L
'letter L/s'

Inanimates: 3/4

d. um-bhobho/imi-bhobho
3-pipe/4-pipe
'pipe/s'

e. um-nqwazi /imi-nqwazi
3-hat/4-hat
'hat/s'

f. um-ba/imi-ba
3-issue/4-issues
'issue/s'

Inanimates: 5/6

g. i-qhosha/ama-qhosha
5-button/6-button
'button/s'

h. i-cepe/ama-cepe
5-spoon/6-spoon
'spoon/s'

i. i-gama/ama-gama
5-word/6-word
'word/s or name/s'

Inanimates: 7/8

j. isi-bane/izi-bane
7-lamp/8-lamp
'lamp/s'

k. isi-tya/izi-tya
7-dish/8-dish
'dish/es'

l. isi-Xhosa
7-Xhosa
'Xhosa language'

Inanimates: 9/10

m. in-to/izin-to
9-thing/10-thing
'thing/s'

n. in-cwadi/iin-cwadi
9-book/10-books
'book/s'

o. in-daba/iin-daba
9-news/10-news
'news item/s'

The generic term for 'tree' is *umthi*, a noun of classes 3/4, as is consistent with the proto-Bantu reconstructions of these classes' contents; many tree terms are of the form *umthi wa X* - 'tree of X', especially those that bear fruit. Specific names for species of trees are often of classes 3/4 as well, but quite a lot fall outside them as exemplified below. The generic term for 'plant' is a deverbal noun of classes 7/8 (*isi-tyalo* < *uku-tyala* -- 'to plant'), and generic 'bush' is a noun of 5/6. I have not (yet) found plant terms in classes 1a/2a.

(16) Plants and trees: 3/4

a. um-thi/imi-thi
3-tree/4-trees
'tree/s'

b. um-hlonyane/ imi-hlonyane
3-Wormwood/4-Wormwood
'Wormwood tree/s'

c. um-nquma/imi-nquma
3-wild.olive/4-wild.olive
'wild olive tree/s'

Plants and trees: 5/6

d. i-khala
5-aloe
'aloe/s'

e. i-tyholo/ama-tyholo
5-bush/6-bushes
'bush/es'

f. i-cuba
5-tobacco
'tobacco'

f. isi-thungwa/izi-thungwa g. isi-tyalo/izi-tyalo h. isi-Qwane/izi-Qwane
7-sew.PASS/8sew.PASS 7-plant.NOM/8-plant.NOM 7-Protea/8-Protea
'Cape Quince' (plaited for use) 'plant/s' 'Protea'

i. in-tshilo/iin-tshilo
9-TSHILO /10-TSHILO
'Hedge Caper'

j. in-yibiba/iin-yiba
9-lily/10-lillies
'lily/lilies'

k. in-gcongolo/iin-gcongolo
9-reed/10-reeds
'reed/reeds'

a. **um**-chamo b. **i**-gazi c. **ama**-nzi d. **isi**-dudu e. **i**-oili f. **u**-bisi
3-urine 5-blood 6-water 7-porridge 9-oil 11-milk
'urine' 'blood' 'water' 'porridge' 'oil' 'milk'

g. **i**-tyuwa h. **um**-gubo i. **ama**-zimba j. **u**-daka
9-salt 3-flour 6-sorghum 11-mud
'salt' 'flour' 'sorghum' 'mud'

(18) Clauses a. Ndi-ya-(**ku**)-cinga ukuba u-hambile uSabelo. b. Uku-cula **ku**-mnandi.
class 15 1SSM-DISJ-15SM-think that 1SM-left 1a-Sabelo 15-sing 15SM-nice
'I think that Sabelo left.' 'Singing is nice.'

(19) a. **um**-fazi **w**-a-fika. b. **in**-tombi **y**-a-fika. c. **i**-gwetha **l**-a-hleka.
 1-woman **1SM**-PST-arrive **9**-girl **9SM**-PST-arrive **5**-lawyer **5SM**-PST-laugh
 ‘The woman arrived.’ ‘The girl arrived.’ ‘The lawyer laughed.’

(20) a. **u**-lolilwe **w**-a-fika. b. **i**-posi **y**-a-fika. c. **ili**-tya **l**-a-wa phantsi.
 1a-train **1SM**-PST-arrive **9**-mail **9SM**-PST-arrive **5**-stone **5SM**-PST-fell down
 ‘The train arrived.’ ‘The mail arrived.’ ‘The stone fell.’

(21) a. **Aba**-fazi/**oo**-lolilwe **ba**-a-fika. b. **lin**-tombi/**iin**-cwadi **z**-a-fika
 2-women/**2a**-trains **2SM**-PST-arrive **10**-girls/**10**-letters **9SM**-PST-arrive
 ‘The women/trains arrived.’ ‘The girls/letters arrived.’

 c. **Ama**-gwetha/**ama**-tye **a**-a-wa. d. **Imi**-thi/**imi**-gewu **y**-a-wa.
 6-lawyers/**6**-stones **6SM**-PST-fell **4**-trees/**4**-criminals **4SM**-PST-fell
 ‘The lawyers/stones fell.’ ‘The trees/criminals fell down.’

11

consistent semantic features apart from a few limited strands of coherence; but as §1 pointed out, agreement with conjoined singulars reveals a conceptual division of [human] vs. others. We see this manifested in the Xhosa (22)a,b where singular nouns from different classes are conjoined. In (22)a where the conjoined nouns denote inanimates, agreement is class 8. For conjoined [human] nouns of mismatching classes agreement is class 2.

- (22) a. Um-nqathe ne-qanda **zi-se** tafile-ni. [Xhosa; Mitchely 2015:115 adapted]
 3-carrot and.5-egg 8SM-LOC table-LOC
 'The carrot and the egg are on the table.'
- b. Um-gulukudu ne-polisa **ba-ya-baleka**
 3-gangster and.5-policeman 2SM-DISJ-run
 'The student and the policeman are running.'

The pattern is common across Bantu languages -- agreement with mismatched conjuncts is quite generally from class 2 for [human] nouns and from class 8 for others, regardless of the conjuncts' intrinsic classes (Corbett 1983, 1991, 2006, Corbett & Mtenje 1987, De Vos & Mitchley 2012, Diercks et al 2015, Givon 1970, Marten 2000, 2005, Roberts & Wolontis 1974, Schadeberg 2001, Simango 2012, and Voeltz 1971, Mitchley 2015). (23) and (24) provide examples from the literature (glosses adapted); (25) is a baseline descriptive generalization.

- (23) a. Mbale na ka-temo **v-a-sowa**. [ciNsenga; Simango 2012: 178]
 9.plate and 12-axe 8SM-past-miss
 'The plate and the axe are missing.'
- b. Li-doda na m-simbi **w-e-fwika** mailo [ciNsenga; Simango 2012:179]
 5-man and 1-girl 2SM-PST-arrive yesterday
 'The man and the girl arrived yesterday.'
- (24) a. U-mu-bíírá ní-í-ri-gena m-**bí-síir:e** [Kuria; from Diercks et al 2015:31]
 3-3-ball and-5-5-stone FOC-8SM-disappear.PST
 'The ball and the stone disappeared.'
- b. Axi-kelema no buchara **vo** vulavula. [Xitsonga; Mitchley 2015:65]
 DEF-7-scoundrel and 9.bucher 2SM.PRES talk
 'The scoundrel and the butcher are talking.'
- (25) Default or semantic agreement rules for Bantu:
 a. For conjoined mismatched [+human] nouns, agreement is class 2.
 b. For conjoined mismatched [-human] nouns, agreement is class 8.

Though I will use the term default agreement for convenience, there are clearly semantic underpinnings to this strategy as has been widely noted in the literature since, as we have seen, classes 1/2 are reconstructed as the canonical domain of [human] nouns and Table 1 characterizes the contents of classes 7/8 as mixed. In fact 7/8 are often defined as the canonical [inanimate] classes, as we will see reflected in discussion of Denny & Creider (1986) in §4.

3.2 Cases where a default strategy is puzzling

3.2.1 Singular inanimates [3+3], [5+5] pair with class 8 *zi-*

As pointed out in Taraldsen et al (2018) and mentioned in §1, some pairs of conjoined singulars drawn from a single class – in particular the combinations [3+3] and [5+5] – also take these same *ba-* and *zi-* defaults in Xhosa. (26)a and (27)a show this for conjoined [inanimate] nouns: agreement with them is class 8 *zi-*, though as (26)b and (27)b demonstrate, the same nouns' plurals belong to class 4 and 6 respectively, and pair with the agreement markers *i-* and *a-*.

- (26) a. Um-nqwazi nom-pu √**zi-** /X**i-** se tafile-ni. [3+3=8; ≠4]
 3-hat and.3-gun 8SM/ 4SM-are table-LOC
 'A hat and a gun are on the table.'
- b. Imi-nqwazi Xzi /√**i-**se tafile-ni. [plural of N_{cl.3} is cl 4: *i-*agr]
 4-hats 8SM/ 4SM-be table-LOC
 'The hats are on the table.'
- (27) a. Ili-tye ne-qanda √**zi-** /X**a-**nyamalele. [5+5=8, ≠ 6]
 5-stone and.5-egg 8SM/ 6SM-disappeared
 'The stone and the egg disappeared.'
- b. Ama-tye Xzi /√**a-**nyamalele. [plural of N_{cl.5} is cl 6: *a-*agr]
 6-stones 8SM/ 6SM-disappeared
 'The stones disappeared.'

3.2.2 Conjoined humans of [3+3], [5+5] pair with class 2 *ba-*

Similarly, conjoined [human] nouns of these classes pair with the class 2 prefix *ba-* rather than with the agreement markers accompanying their plurals ((28)a,b and (29)a,b).

- (28) a. Um-gewu nom-gulukudu √**ba** /X**i-**sebenza ndawonye. [3+3=2, ≠4]
 3-criminal and.3-gangster 2SM/ 4SM-work together
 'A criminal and a gangster are working together.'

- b. I-mi-gewu X**ba**/√**i**-sebenza ndawonye.
 4-criminals 2SM/ 4SM -work together
 'The criminals work together.'
- (29) a. I-gqirha ne-gosa √**ba** /X**a**-sebenza ndawonye. [5+5=2, ≠ 6]
 5-healer and 5-officer 2SM/ 6SM-work together
 'The healer and the officer are working together.'
- b. A-ma-gqirha X**ba**/√**a**-sebenza ndawonye.
 6-healers 2SM/ 6SM-work together
 'The healers are working together.'

Taraldsen et al (2018) conclude from this that Bantu singular/plural pairings do not share gender features. Each singular and each plural class is a distinct gender.

3.3 Analyses of parallel facts in canonical gender systems.

Contrary to Taraldsen et al's claim, the patterns of default agreement in (26)-(29) do not threaten the gender analysis of pairs of classes owing to well-known parallels in languages with more canonical three-gender systems. As noted in §1, neuter singulars in BCS do not pair with plural neuter agreement, but rather with default masculine as shown in (5) and in (30) below:

- (30) *Jedno tele i jedno pašče su juče prodana. [Bošković 2009]
 one calf.neut and one dog.neut are yesterday sold.pl.neut
 Intended: a calf and a dog were sold yesterday.'

Marušić et al (2007) and Bošković (2009) propose an account: Conjunct Phrase (&P) has number features only, and it's a closer goal to a clause-level agreement probe than are &P's contents. &P therefore intervenes to block access to the conjuncts' features. A uPhi probe on a head H obtains from &P a plural value alone, leaving the linked *ugender* feature unvalued.

Default/semantic resolution rules follow whether the genders of conjuncts mismatch or match.

- (31) a. b.

3.4 Where regular plural agreement obtains

3.4.1 Conjoined singulars [1+1], [7+7], [9+9] pair with expected plural agreement

There is a drawback to (31): the approach suggests that only default agreement should be possible with conjoined singulars. Yet as I mentioned in the introduction, agreement *does* match, in certain genders. In Xhosa, conjoined singular nouns of class 1 take class 2 *ba*-AGR (see (32) and (33)). This is true whether or not they are [human], arguing that the pattern cannot be attributed to their intrinsic semantics triggering the default rule in (25)a.

- (32) Um-mi nom-ongameli **ba**-ya-ncokola. [1+1=2]
 1-citizen and.1-president 2SM-DISJ-chat
 'The citizen and the president are chatting.'
- (33) a. U-L no-M **ba**- /*zi-se tafile-ni. [1+1=2]
 1a-L and.1a-M 2SM-/*8SM-LOC table-LOC
 'The L and the M are on the table.'
- b. U-loliwe kunye no-matshini **ba**-/*zi-ya-hamba. [1+1=2]
 1a-train and and.1a-machine 2SM-/*8SM-DISJ-move
 'The train and the machine are moving.'

Similarly, conjoined singulars of class 7 take class 8 *zi*-agr whatever they denote; thus conjoined [human] nouns in class 7 do not pair with the class 2 *ba*- that serves as default for conjunctions of [human] nouns from classes [3+3] and [5+5].

- (34) Isi-bane nesi-tya **zi**-nyamalele. [7+7=8]
 7-lamp and.7-dish 8SM-disappeared
 'The lamp and the dish have disappeared.'
- (35) Is-anuse nes-azi **zi**-ya-sebenza.
 7-diviner and.7-scientist 8SM-DISJ-work
 'The diviner and the scientist are working.'

Conjoined singulars of class 9 also take the *zi*-AGR, even if [human]:

- (36) In-dlovu (kunye) nen-gwe **zi**-ya-lwa. [9+9=10]
 9-elephant and and.9-leopard 8SM-DISJ-fight.
 'The elephant and the leopard are fighting.'
- (37) In-dadi nen-tlebi **zi**-ya-cula.
 9-swimmer and.9-gossip 10SM-DISJ-sing
 'The swimmer and the gossip are singing.'

Class 8 and 10 agreement morphology is syncretic; *zi*- is ambiguous between the two classes. I analyze the *zi*- accompanying conjoined class 9 singulars as class 10 rather than default class 8 since it pairs with conjunctions of [human] nouns like those in (37), for which the default is not

zi- but rather *ba-*. In other words this is formal grammatical agreement, like systematic gender-matching agreement with all [1+1] and [8+8] conjunctions as exemplified in (32)-(35).

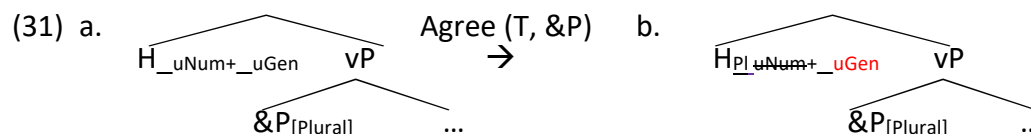
3.4.2 More parallels in BCS

This pattern too has parallels outside of Bantu: as we have seen, though BCS [_{&P} neut+neut] take default masculine, [_{&P} fem+fem] and [_{&P} masc+masc] can take matching agr ((4) from Murphy & Puskar 2018; glosses adapted). Since masculine is the BCS default, the implications of (38) are not clear, but feminine grammatical agreement clearly succeeds with &P in cases like (4) whereas we saw in (5) and (30) that neuter fails in the same syntactic circumstance.

(38) Ovaj covek i onaj decak su stigl-i. [BCS: Despić 2017: 291]
 this man.MASC and that boy.MASC are arrived. MASC.PL
 'This man and that boy arrived.'
 [_{&P} MASC + MASC = MASC.PL]

(4) [_{&P} Zavesa i biljka] su ukrašavale prozor.
 curtain.F.SG and plant.F.SG are decorate.PRT.F.PL window
 'A curtain and a plant decorate the window.'
 [_{&P} FEM + FEM = FEM.PL]

The facts are at odds with proposal (31) (repeated below).



Some recent works address the contrast between BCS neuter and the other genders differently: Despić (2016), Nevins (2018), Tsimplici & Hulk (2013), propose that gender agreement is not blocked in this configuration but fails for conjoined neuters because neuter is the absence of gender or negative values [-FEM, -MASC]. Adamson & Anagnostopoulou (2024) follow Arsenijević (2017) in relating the facts to the status of BCS neuter as intrinsically [mass], hence non-individuated and largely incompatible with regular pluralization. Such approaches seem reasonable for languages with a single outlier gender which exhibits these or similar anomalies. But the ideas do not extend well to Xhosa where there are two outliers, and as we

will see in §7, Shona has six; plural formation in these genders is regular and their semantic contents are heterogeneous.

Bošković (2009) suggests that BCS feminine but not neuter is reflected in agreement with conjoined singulars because feminine is semantically grounded and therefore percolates to &P, though why this should be so in cases like (4) he leaves open. Adamson & Anagnostopoulou (2024) also articulate an approach to the contrast in terms of gender semantics. I'll build on these proposals an analysis based on the semantics of genders and *nP* stacking (Kramer 2015).

4. Analysis

4.1 Why 1/2, 7/8, and 9/10 are different from 3/4, 5/6

I summarize in (39) the Xhosa patterns to be captured:

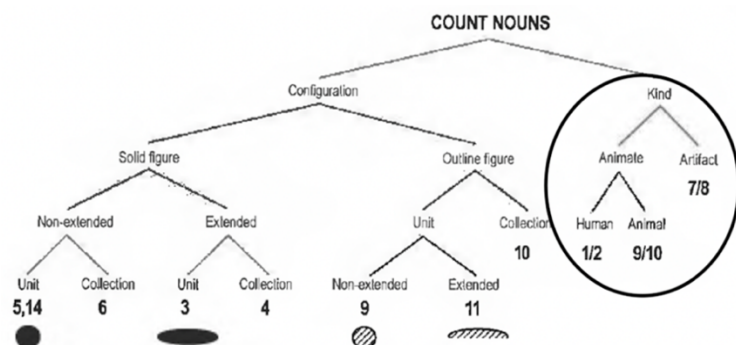
- (39) a. Conjoined singulars [1+1] [7+7], [9+9] control matching plural agreement.
b. [3+3] and [5+5] take default agreement.
c. Classes 2 and 8 (plurals of 1 and 7) provide default agreement for [3+3], [5+5] and for mismatching [X_{sing} & Y_{sing}].

Thus two of the genders exhibiting grammatical agreement with conjoined, gender-matched singulars are also the sources of default plural agreements -- classes 1/2, 7/8 -- and the defaults correlate with semantic associations of classes 1/2 to [human], and 7/8 to non-human. By logical extension, the pattern [9+9=10] suggests that 9/10 have a comparable semantic association; I assume that 9/10 is [animal] based on the reconstructed contents of 9/10 as well as its inclusion of many animal names synchronically; 7/8 is [inanimate]. This makes 1/2, 7/8, and 9/10 a grouping of Bantu genders with recognized semantic connections to clearly defined classes of entities. It's puzzling, though, that 3/4 are not part of this grouping, given their standard reconstruction as the classes of trees and plants.

There is a basis for positing that the semantics of this gender are somewhat more

amorphous and, building on recent approaches to neuter, that this might be at fault. Denny & Creider (1986) propose that only proto-Bantu 1/2, 7/8, and 9/10 have semantic associations to entity types; 3/4, 5/6 and others are instead characterized by abstract shape tendencies. I reproduce their schematic for this in Figure 1 in which I circle Denny & Creider's 'kind' node and its dependents. Denny & Creider argue that the number of reconstructed tree and plant terms in proto-Bantu classes 3/4 is statistically insufficient to justify their inclusion beneath this node.⁴

Figure 1: Proto-Bantu noun class semantics in Denny & Creider (1986)



A semantics of shape rather than entity type might lead us to suspect that 3/4 and 5/6 share with BCS neuter a sort of defectiveness precluding plural agreement with conjoined singulars. However §7.1 will show that nouns of classes 3/4 and 5/6 are entirely regular according to a range of diagnostics that BCS neuters fail, and §7.2 will present evidence that in Shona, where the number of genders is larger, conjoined singulars of *most* genders cannot pair with regular plural agreement – in other words, 3/4 and 5/6 are not outliers at all, but perhaps instantiate the broadest pattern where agreement with conjuncts is concerned. Assuming this, then what is needed is a characterization of genders for which agreement with conjuncts *does*

⁴ Xhosa botany: Bhatt (2013) lists 30 plants and trees of which 8 are in classes 3/4 = 26%; Bhatt (2014) lists 11/29 in class 3/4 -- 61%. In Wehmeyer & Rose (1983) plants and trees in classes 3/4 number 3/18 or 17%. On the other hand, Muluwa (2010) found nearly 50% of plant names in 3/4 in a set of Congolese languages.

succeed rather than of the genders for which it does not. It is accordingly quite possible that 3/4 do have some semantic association to trees and plants, but that this is immaterial.

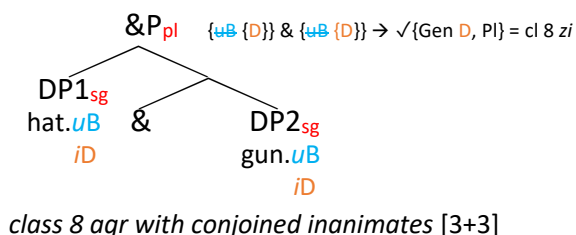
I suggest that a gender with recognized association to a culturally salient group of entities like [male], [female], [human], [animal], [artifact] may have formal legibility in the syntax in a way that other genders don't. I will classify the former type of gender as having interpretable content in the syntax of the relevant language, and the others as uninterpretable. It seems clear that what is or is not adopted as interpretable varies across languages somewhat arbitrarily.

Returning to the issues at hand, I analyze 3/4 and 5/6 as wholly *ugenders*; this is why they do not figure into resolved agreement which, following Adamson & Anagnostopoulou (2024), is based on *i*features alone as will be detailed in §4.3. (40) illustrates the layered structure of a class 3/4 noun, and (42) previews how default agreement in (41) is derived via deletion of *ugenders* on &P. We will see evidence that gender features of each conjunct DP are nested in accordance with the heights of their *n*P sources, as the curly bracketings in (42) indicate.

(40) um-qwazi/imi-qwazi [*n*_B [*n*_D √QWAZI]] *un* of 3/4 stacks above 7/8
 3-hat/4-hat
 'hat/s'

(41) a. um-nqwazi nom-pu √zi- /X*i*- se tafile-ni. [3+3=8; ≠4]
 3-hat and.3-gun 8*SM*/ 4*SM*-are table-LOC
 'A hat and a gun are on the table.'

(42) singular conjuncts trigger the resolution process



4.2 Gender, *n*, and interpretability

Following Kramer (2015) I take gender to be a feature of the categorizer *n* as shown in (43):

- (43) The categorizers:
- a. vP

```

graph TD
    vP --> v
    vP --> vP2[vP]
    vP2 --> hammer

```

b. nP

```

graph TD
    nP --> n
    nP --> nP2[nP]
    nP2 --> hammer

```

Kramer proposes that genders may have *i*- and *u*- “flavors”. Amharic has two genders: feminine and masculine, with *i*flavors [+FEM] and [-FEM] for female- and male-denoting nouns respectively. Nouns that are arbitrarily feminine have *u*[+FEM] gender, and the remainder are masculine, by default. I summarize in (44) and illustrate in (45) with two Amharic nouns naming entities with natural gender and with the arbitrarily feminine noun ‘sun’:

(44) Amharic: two genders. Types of *n*:

- a. *n i* [+FEM] Female natural gender
- b. *n i* [-FEM] Male natural gender
- c. *n u* [+FEM] Feminine arbitrary gender
- d. *n* No natural gender = “plain” *n* (grammatically masculine, by default)

- (45)
- a. nP

```

graph TD
    nP --> vP
    nP --> ni["n i [+FEM]"]
    vP --> vHAKIM
    vHAKIM --> femdoc["'female doctor'"]

```

b. nP

```

graph TD
    nP --> vP
    nP --> ni["n i [-FEM]"]
    vP --> vHAKIM
    vHAKIM --> maledoc["'male doctor'"]

```

c. nP

```

graph TD
    nP --> vP
    nP --> nu["n u [+FEM]"]
    vP --> vS_AHAY["vS'ÄHAY"]
    vS_AHAY --> sun["'sun' (arbitrary feminine)"]

```

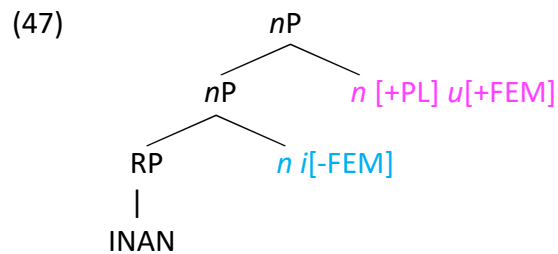
I adopt the little *n* approach, the proposal of *u*- and *i*gender varieties, and a second tool from Kramer: gender stacking.⁵ Kramer introduces this to account for plural gender-shift in Somali. She proposes that *inan* ‘son’ or ‘boy’ shifts from masculine to feminine when it pluralizes because Somali includes a plural *n* with the gender feature *u*[+FEM]. When *inan* pluralizes, this *n* stacks above the interpretable [-FEM] *n* to create a plural that is grammatically feminine though semantically masculine, as shown in (47).

- (46)
- a. *inan* 'son, boy (m.)'

b. *inammó* 'sons, boys (f.)'

[Somali]

⁵ See Carstens (2019) for a first sketch of this approach to Xhosa default/semantic agreement, adopted in Carstens’s (2023, 2024a,b) presentations pursuing a Multiple Agree implementation to both conjoined singulars and conjoined plurals, and Carstens (2024) for a precursor to the present analysis. See also Fuchs & van der Wal (2022) for analysis of Bantu diminutives and locatives in terms of *ns* and *nP*-stacking.



Returning to Xhosa, the phenomena discussed suggest the following generalizations:

- (48) a. wholly uninterpretable genders are recognizable as such throughout the derivation.
 b. *u*- and *i*flavors of a single gender are not distinguished in the morphosyntax.

In Kramer's account, licensing conditions ensure that combinations of [gender+root] are compatible. For any *u*flavor, the relevant licensing condition applies at PF. This meshes with the evidence that Xhosa's *ugenders* 3/4 and 5/6 never factor into the calculus of feature-resolution on &P, giving rise to "default" agreement with conjoined singulars. We need only assume that wholly uninterpretable gender features are deleted from (or irrelevant to) the syntactic representation in the resolution process, though their PF features remain.

Kramer proposes that only the licensing conditions for [*i*gender+root] apply at LF. To capture the success of appropriate plural agreement with all conjunctions [1+1], [7+7] and [9+9], I depart from this in proposing instead that any [gender+root] combination involving a gender that has an *i*flavor is subject to LF licensing conditions. This seems a natural means to ensure that flavors of the same gender are properly distributed.

Recall the Bantu gender system proposed in Carstens (1991, 2008), reproduced below.

- (10) Bantu Genders (Carstens 1991)

Gender **A**: stems of Classes 1/2
 Gender **B**: stems of Classes 3/4
 Gender **C**: stems of Classes 5/6

Gender **D**: stems of Classes 7/8
 Gender **E**: stems of Classes 9/10
 Etc.

(49) provides an inventory of *ns* for Xhosa based on the properties of the genders. Genders A

and D, underlying classes 1/2 and classes 7/8, have flavors *i*[human] and *i*[inanimate] respectively, accounting for two things: their survival in agreement with matched singular conjuncts [1+1] and [7+7], and their use as defaults for conjunctions of singular human and inanimate-denoting nouns. I have posited an *i*[animal] flavor for Gender E of classes 9/10 to account for class 10 agreement with conjoined singulars of class 9. Genders B and C, underlying 3/4 and 5/6 respectively, lack interpretable flavors as shown in (49). For expository clarity I identify flavors other than the canonical semantic content of a gender as *arb* to distinguish them from the wholly *ugenders* 3/4 and 5/6, though 3/4 and 5/6 are in fact always arbitrary, and the flavors A2, D2, and E2 are uninterpretable though this is opaque in narrow syntax.

(49) ns for Xhosa, reflecting conjunct agreement facts

Classes 1/2 = Gender A	<i>n</i> _{A1}	<i>i</i> [human]
	<i>n</i> _{A2}	<i>arb</i>
Classes 3/4 = Gender B	<i>n</i> _B	uninterpretable, for all members
Classes 5/6 = Gender C	<i>n</i> _C	uninterpretable, for all members
Classes 7/8 = Gender D	<i>n</i> _{D1}	<i>i</i> [artifact]
	<i>n</i> _{D2}	<i>arb</i>
Classes 9/10 = Gender E	<i>n</i> _{E1}	<i>i</i> [animal]
	<i>n</i> _{E2}	<i>arb</i>

Comparing the synchronic contents of Xhosa classes to the Proto-Bantu reconstructions in Tables 1 and 2, it seems that large numbers of [human] nouns drifted from Proto-Bantu classes 1/2 into other classes. Let's assume their new genders stack above an *icore* of Gender A.

(50) Sample structures of [human] nouns: a core of 1/2 = gender A

- a. *um*-ntwana/aba-ntwana [*n*_{A1} √MNTWANA] *n*_A for [human]
1-child/2-children
'child/ren'
- b. *um*-gewu/*imi*-gewu [*n*_B [*n*_{A1} √GEWU]] *un*_B of 3/4 stacks above 1/2 *n*_A
3-criminal/4-criminal
'criminal/s'
- c. *i*-butho/*ama*-butho [*n*_C [*n*_{A1} √BUTHO]] *un*_C of 5/6 stacks above 1/2 *n*_{A1}
5-warrior/6-warrior
'warrior/s'

- d. isi-hlobo/izi-hlobo [n_{D2} [n_{A1} VHLOBO]] n_{D2} of 7/8 stacks above 1/2 n_{A1}
 7-friend/8-friend
 'friend/s'
- e. in-tombi/iin-tombi [n_{E2} [n_{A1} VTOMBI]] n_{E2} of 9/10 stacks above 1/2 n_{A1}
 9-young.lady/10-young.lady
 'young lady/ladies'

Similarly, I propose that [inanimate] nouns have class 7/8 cores. If their synchronic class membership is different, a higher *n* introducing an outer gender is layered above this:

(51) Structures of [inanimate] nouns: a core of 7/8 = gender D

- a. isi-bane/izi-bane [n_{D1} VBANE] n_{D1} for [artifact]
 7-lamp/8-lamp
 'lamp/s'
- b. u-matshini/oo-matshini [n_{A2} [n_D VMATSHINI]] n_{A2} of 1/2 above n_{D1} of 7/8
 1a-machine/2a-machine
 'machine/s'
- c. um-qwazi/imi-qwazi [n_B [n_D VQWAZI]] un of 3/4 stacks above n_{D1} 7/8
 3-hat/4-hat
 'hat/s'
- d. ili-tye/ama-tye [n_C [n_D VTYE]] un of 5/6 stacks above n_{D1} of 7/8
 5-stone/6-stone
 'stone/s'

[Animal] nouns outside of classes 9/10 have *nP* layers above a gender E core as in (52). I

assume that default *zi*-agreement with [animal] nouns of all classes is of class 10, though syncretism with class 8 obscures this.

(52) Structure of [animal] nouns: an inner layer of 9/10 = gender E

- a. in-dlovu/iin-dlovu [n_{E1} VDLOVU] n_{E1} for [animal]
 9-elephant/10-elephant
 'elephant/s'
- b. u-nokala/oo-nokala [n_{A2} [n_E VNOKALA]] n_{A2} of 1/2 above n_{E1} of 9/10
 1a-crab/2a-crab
 'crab/s'
- c. um-vundla/imi-vundla [n_B [n_E VQAZI]] un of 3/4 above n_{E1} of 9/10
 3-rabbit/4-rabbit
 'rabbit/s'
- d. isi-khova/izi-khova [n_D [n_E VQAZI]] n_{D2} of 7/8 above n_{E1} of 9/10
 7-owl/8-owl
 'owl/s'

5 Deriving the patterns

5.1 Xhosa conjoined singulars

I turn to the derivation of agreement with &P. In a study of Chichewa gender, Corbett & Mtenje (1987) propose that conjunctions of singulars and conjunctions with gender mismatch are both triggers for gender resolution. Corbett & Mtenje show that Chichewa [3+3] and [5+5] pair with default class 2 agreement for [human] and with syncretic 8/10 agreement for other semantic content just as we have seen in Xhosa. They propose that resolution for number forces resolution for gender. If conjuncts are plural and match in gender, resolution does not apply.

The correct generalization appears to be that gender resolution can be triggered in two ways. First it must operate if number resolution operates. There is a very general principle that if one resolution rule operates, all must operate if possible. This solution requires a stipulation in the number resolution rule that number resolution can operate only provided there is at least one singular conjunct. The second possible trigger for gender resolution is the presence of different genders in the subject....It followed as a logical consequence of this analysis that examples with conjoined masculine or conjoined feminine plurals ...do not undergo resolution... (Corbett & Mtenje 1987: 24)

That singular but not plural number triggers resolution in Xhosa is confirmed by comparison of gender-matching singulars of 3/4 and 5/6 in e.g. (53) versus their plurals in (54); the latter pair with gender-matching plural agreement, unlike their conjoined singular counterparts.

- (53) a. Um-nqwazi nom-pu **zi**-se tafile-ni.
 3-hat and.3-gun 8SM-are table-LOC
 'A hat and a gun are on the table.'
- b. I-gqirha ne-gosa **ba**-ya-sebenza.
 5-healer and.5-officer 2SM-DISJ-work
 'The healer and the officer are working.'
- (54) a. Imi-pu nemi-nqwazi **i**-se tafile-ni.
 4-guns and.4-hats 4SM-are table-LOC
 'The guns and the hats are in the room.'
- b. Ama-gqirha nama-gosa **a**-ya-sebenza.
 6-healers and.6-officers 6SM-DISJ-work
 'The healers and the officers are working.'

I turn to the mechanics of resolution for conjoined singulars. I adopt for Xhosa the proposal

of Marušić et al (2007) and Bošković (2009) that &P is plural; the mismatch between &P and singular conjuncts forces the resolution process. Following Adamson & Anagnostopoulou (2024), features of conjuncts *percolate* to &P and resolved agreement is based on a *conversion* which intersects their *i*-features to form a single set comprised of shared values. For Adamson & Anagnostopoulou this is based on feature-dependencies such as (55), for Greek.

(55) Greek gender in Adamson & Anagnostopoulou (2024): (*i/u*) CLASS = entity (neuter)

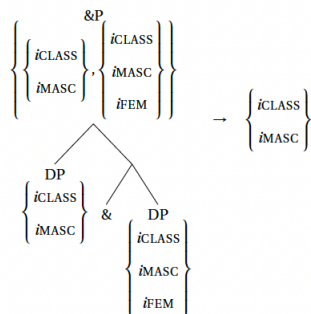
|
(*i/u*) MASC = animate
|
(*i/u*) FEM = woman/women

(57) illustrates the conversion process that derives agreement with the mismatched Greek conjuncts in (56)a. Their masculine and feminine features are interpretable because these are [animate] conjuncts. Based on the dependencies in (55), the conjunction in (56)a resolves to masculine. In contrast the masculine and feminine features of (56)b are not interpretable, so they do not figure into conversion and the intersection is [class], realized in Greek as neuter.⁶

- (56) a. O andras ke i gineka ine eksipni /*eksipna. [Greek]
the.M.SG man and the.F.SG woman are intelligent.M.PL/ *intelligent.N.PL
'The man and the woman are intelligent.'
- b. O pinakas ke i karekla ine vromika /*vromiki.
the.M.SG blackboard and the.F.SG chair are dirty.N.PL/ *dirty. M.PL
'The blackboard and the chair are dirty.'

(57) Deriving masculine agreement in (56)a:

Mismatch:



⁶ I address an incompatibility between this approach to (56)b and the account I am presenting below.

We do not find anything quite like the relationships in (55) among the relevant features in Xhosa. What I've analyzed as outer gender shells are not featurally more specific than the [human], [inanimate], [animal] genders that serve as defaults when conjuncts' genders mismatch or when nouns of classes 3/4 or 5/6 are conjoined. Nor is there evidence of nesting dependency relations among these three semantic cores in Xhosa, which seem instead to function as opposing and co-equal primitives.⁷ The approach that I advocate here in terms of Bantu *nP*-stacking is accordingly a bit different from Adamson & Anagnostopoulou's, but their proposals of percolation and resolution to shared features provide valuable tools and insights nonetheless.

(58) Number and gender resolution triggered by [sing]+[sing].

class 8 agr with conjoined inanimates [3+3]; = (53)a

class 2 agr with conjoined humans [5+5] = (53)b

⁷ In Swahili all animal nouns whatever their singular/plural patterns combine with agreement of the prototypically [human] classes 1/2, diagnosing 1/2 as redefined as [animate] gender. Paralleling the Greek dependency relations sketched in (55) we might expect some Bantu languages to have shifted humans from classes 1/2 into 9/10, the reconstructed classes of animals and therefore suitable for expansion to animate generally. I have not encountered such a pattern but leave this to be explored in future cross-Bantu research.

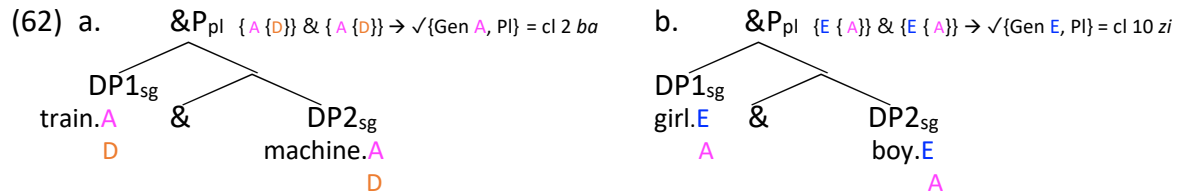
Adamson & Anagnostopoulou (2024) propose that matching features, including uFs, do not trigger conversion. This is why the feminine conjuncts of ‘the skirt and the t-shirt’ pair with gender-matching plural agreement in (59) even though they bear arbitrary *ugender*; that the same does not happen for [neuter+neuter] in BCS is due to neuter’s exceptionality.

- (59) I fusta ke i bluza ine vromikes. [Greek; Adamson & Anagnostopoulou 2024:39]
 the.F.SG skirt and the.F.SG t-shirt are dirty.F.PL
 ‘The skirt and the t-shirt are dirty.’

The approach is not compatible with the failure of matching agreement with Xhosa [3+3] and [5+5] as discussed above, or in several additional classes in Shona to be exemplified in §7.2. As already noted, these genders pattern with BCS neuter; the facts argue that the latter is less exceptional than it has heretofore seemed. Instead I treat BCS neuter and Xhosa classes 3/4 and 5/6 as purely *ugenders* excluded from the resolution process as sketched in (58).

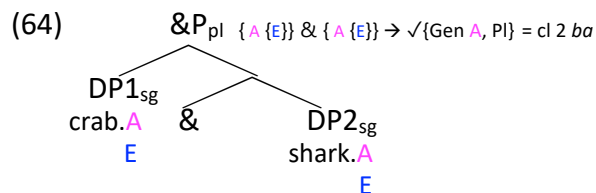
In cases like (60) and (61), matching arbitrary gender flavors stack above matching *igender* cores. Recall my proposal that arbitrary flavors are not distinguished from their *icounterparts* of the same gender before inspection at LF; they therefore factor into resolution. The arbitrary genders match and are highest in (60) and (61) as the nested sets in (63) demonstrate, so they “win” (Kramer 2015, and see Carstens’s 2010, 2011 proposal that N-to-D gives all probes outside DP full access to Bantu gender features). This is a first indication that features are percolated in hierarchies according to their *nP* sources and that that this impacts resolution.

- (60) U-loliwe kunye no-matshini **ba-** /*zi-ya-hamba. [1a+1a=2]
 1a-train and and.1a-machine 2SM-/*8SM-DISJ-move
 ‘The train and the machine are moving.’
- (61) In-tombazana nen-kwenkwe **zi-** /*ba-ya-sebenza. [9+9=10]
 9-girl and.9-boy 10SM/*2SM-DISJ-work
 ‘The girl and the boy are working.’



Conjunctions of [animal] nouns follow the same patterns. If both are drawn from gender A, they pair with class 2 plural *ba*-agreement of gender A though it is, in this instance, arbitrary.

- (63) U-nokala no-krebe **ba**-ya-lwa
 1a-crab and.1a-shark 2SM-DISJ-fight
 'The crab and the shark are fighting.'

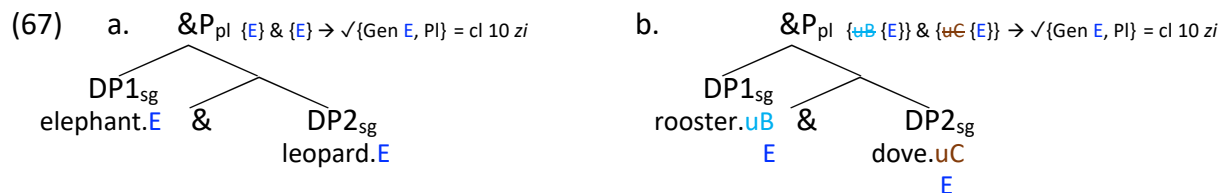


Class 9 [animal] nouns such as those in (65) have single-layer *nP* structures and when conjoined, we have seen that they take class 10 *zi*-agreement; this is derived as shown in (67)a.

(67)b illustrates how gender mismatched [animal] conjuncts of the *ugenders* such as those in (66) also combine with class 10 *zi*-agreement, based on their gender E cores.

- (65) In-dlovu (kunye) nen-gwe **zi**-ya-lwa. [9+9=10]
 9-elephant and and.9-leopard 10SM-DISJ-fight.
 'The elephant and the leopard are fighting.'

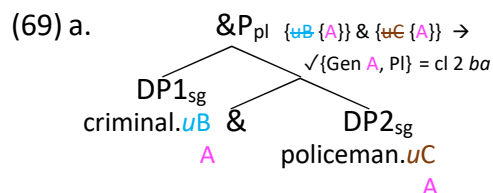
- (66) Um-qhagi ne-hobe zi-haphazela kunye.
 3-rooster and.5-dove 10SM-fly together
 'The rooster and the dove are flying together.'



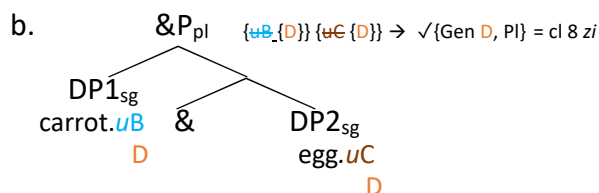
(68)a,b involve singular [human] and [inanimate] nominals respectively, with mismatching *ugenders*. Underlying these genders in each instance there are shared cores of gender A or D as shown in (69)a,b. The cores value agreement once exposed by deletion of B and C.

(68) a. Um-gewu ne-polisa **ba**-sebenza ndawonye. [Xhosa]
 3-criminal and.5-policeman 2SM-pres-work together
 'The criminal and the policeman are working together.'

b. Um-nqathe ne-qanda **zi**-se tatile-ni. [Xhosa; Mitchely 2015:115]
 3-carrot and.5-egg 8SM-loc table-loc
 'The carrot and the egg are on the table.'



class 2 agr with conjoined humans [3+5] = (68)a

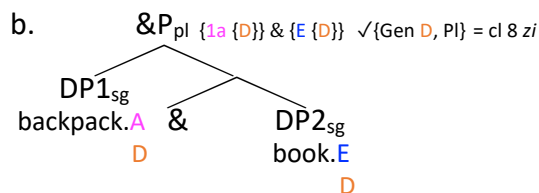
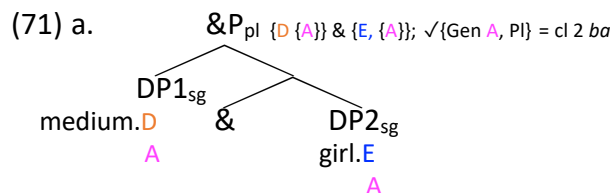


class 8 agr with conjoined inanimates [3+5] = (68)b

The outer genders are also mismatching in examples (70)a,b below. Since these are genders with *iflavors* they are not systematically deleted, making the cases different from those in (68)a,b. But the commonalities are the [human] and [inanimate] gender *icores* respectively, which value agreement. Use of default class 2 *ba*- in (70)a is consistent with the hypothesis that 7/8 and 9/10 are distinct genders even though class 8 and 10 agreements are identical *zi*-.

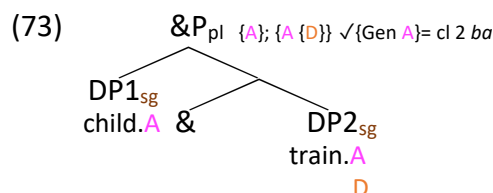
(70) a. Is-anuse nen-tombi **ba**-ya-sebenza.
 7-medium and.9-girl 2SM-DISJ-work
 'The medium and the girl are working.'

b. U-bhaka nen-cwadi **zi**-ngaphandle.
 1a-backpack and.9-book 8SM-be.outside
 'The backpack and the book are outside.'



In (72), the conjuncts share outer gender A which has an *iflavor*, though 'train' has a distinct gender D *icore*. Agreement in the shared outer gender succeeds as illustrated in (73).

(72) Um-twana no-lolilwe **ba**-ya-hamba
 1-child and.1a-train 2SM-DISJ-move
 'The child and the train are moving.'

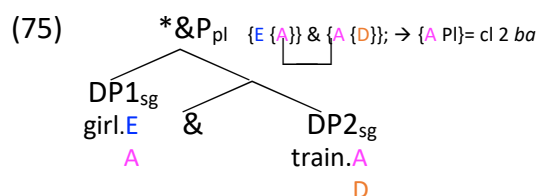


Agreement with conjunctions of humans and inanimates is ineffable if they are drawn from

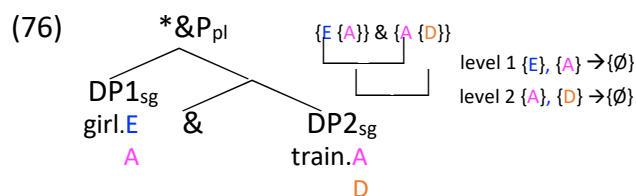
different genders. No choice of resolved agreement works in the following; a paraphrase like ‘The girl collided with the train’ is preferred.

- (74) *In-tombi no-lolilwe zi/ba-gilana.
 9-girl and.1a-train 8SM-/2SM-collide
 'The girl and the train collided.'

Just as in (70) the higher *ns* are mismatching and arbitrary, but in this case, no shared *icore* exists. Yet there is a potential snag: since the *icore* for ‘girl’ is gender A, and the higher gender of ‘train’ is also gender A, a question arises as to why gender A is not identified as the intersection.

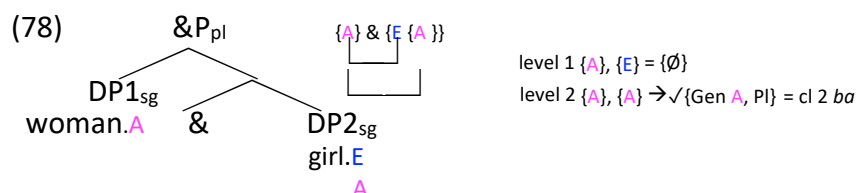


The pattern can be captured by a derivation in which intersection is attempted on the highest feature set and when it fails, the lower genders are considered. As they also mismatch, conversion fails ((76) illustrates).



On the other hand, (77) shows that when two singular [human] nouns are conjoined and one is gender A, class 2 (gender A plural) agreement succeeds. This indicates that the sole gender feature of class 1 ‘woman’ is not excluded from consideration based on mismatching the higher gender E of class 9 ‘girl’. The difference between this case and that in (75)/(76) is that in (77) there is no lower *n* in the first conjunct to be considered in place of gender A of level 1; instead, gender A of DP1 is paired once again, this time with gender A of DP2.

(77) Um-fazi nen-tombazana ba-ya-dla.
 1-woman and.9-girl 2SM-DISJ-eat
 'The woman and the girl are eating.'



The facts confirm that hierarchical factors matter. In (76) when shared outer genders do not match and consideration moves to the lower gender features, these cannot be other than the core for each conjunct; the higher genders are safely disregarded. But in (78), the sole layer of DP1 is the only choice there is to consider; it is the core and matches the core of DP2. The problem in (76) is that the genders at the core level do not match.

5.2 Arbitrary gender revisited: comparing Xhosa and Greek

5.2.1 Xhosa

Let us compare the results that would follow if the flavors of genders that combine with arbitrary members were simply uninterpretable in the syntax, patterning like classes 3/4 and 5/6 instead of with the interpretable flavors. The *u-* and *i-* designations in (79) encode the hypothesis that these values are distinguished throughout the derivation.

(79) Rejected alternative: *u*flavors for 1/2, 7/8, 9/10, adapting Kramer (2015)

Classes 1/2 = Gender A	<i>n</i> _{A1}	<i>i</i> [human]
	<i>un</i> _{A2}	for arbitrary members
Classes 3/4 = Gender B	<i>un</i> _B	uninterpretable, for all members
Classes 5/6 = Gender C	<i>un</i> _C	uninterpretable, for all members
Classes 7/8 = Gender D	<i>n</i> _{D1}	<i>i</i> [artifact]
	<i>un</i> _{D2}	for arbitrary members
Classes 9/10 = Gender E	<i>n</i> _{E1}	<i>i</i> [animal]
	<i>un</i> _{E2}	for arbitrary members

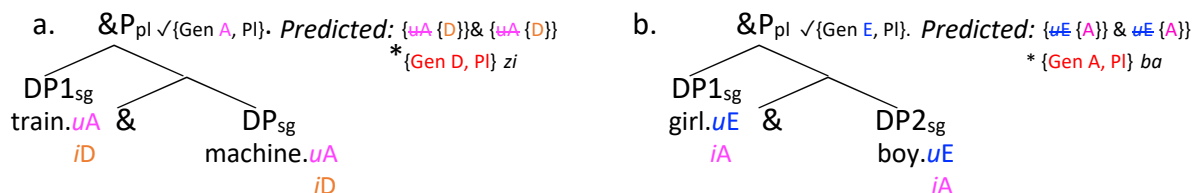
This approach cannot capture such examples as (60) and (61), repeated below. In (60), the nouns for 'train' and 'machine' of classes 1/2 (gender A) are conjoined. Semantically they are

inanimates, and therefore we expect their cores to be gender D = classes 7/8. We would predict that their outer gender A feature would delete if its status as uninterpretable were transparent in syntax, and their gender D feature would be the outcome of conversion, yielding *zi*-agreement. Similarly a conjunction of [human] nouns like those in (61) from a class other than class 1 should take class 2 *ba*- agreement as this is the plural of human "default" i-gender A. As noted above these outcomes would parallel the results of conjoining singular nouns of classes 3 or 5 (examples (28)a and (29)a reproduced in (81) and (82)).

(60) U-loliwe kunye no-matshini **ba-** /**zi*-ya-hamba. [1a+1a=2]
 1a-train and and.1a-machine 2SM/*8SM-DISJ-move
 'The train and the machine are moving.'

(61) In-tombazana nen-kwenkwe **zi-** /**ba*-ya-sebenza. [9+9=10]
 9-girl and. 9-boy 10SM/*2SM DISJ-work
 'The girl and the boy are working.'

(80) *Predictions if u- and i-flavors of the same genders were distinguished in the morphosyntax*



(81) Um-gewu nom-gulukudu √**ba** /X*i*-sebenza ndawonye. [3+3=2, ≠4]
 3-criminal and.3-gangster 2SM/ 4SM-work together
 'A criminal and a gangster are working together.'

(82) I-gqirha ne-gosa √**ba** /X*a*-sebenza ndawonye. [5+5=2, ≠ 6]
 5-healer and.5-officer 2SM/ 6SM-work together
 'The healer and the officer are working together.'

I conclude that arbitrary members of a gender do not pattern as this hypothesis would predict. Instead, all "flavors" of a gender pattern alike with respect to conversion and resolution. This motivates my proposal that if a gender has any interpretable content, both its *i*- and *u*-flavors are retained throughout the syntactic derivation, while in contrast, wholly *u*-genders delete. The sole alternative is to designate genders C and D as non-individuable or

exceptional and defective in some way, adapting the analysis of neuter in Arsenijević (2017, 2021). As previously noted, §7 will detail evidence against this.

5.2.2 Greek

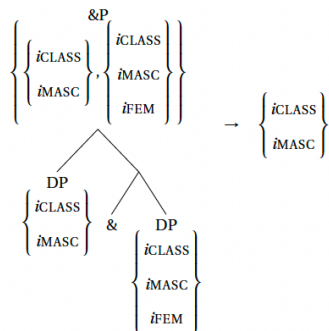
In light of my proposals for Xhosa, let us revisit Adamson & Anagnostopoulou's (56), reproduced below.

- (56) a. O andras ke i gineka ine eksipni /*eksipna. [Greek]
the.M.SG man and the.F.SG woman are intelligent.M.PL/ *intelligent.N.PL
'The man and the woman are intelligent.'
- b. O pinakas ke i karekla ine vromika /*vromiki.
the.M.SG blackboard and the.F.SG chair are dirty.N.PL /dirty. M.PL
'The blackboard and the chair are dirty.'

Recall that for Adamson & Anagnostopoulou, both conjuncts of (56)a include *i*genders. As the meaning of Greek *i*masculine is [animate], and *i*feminine is a dependent of [animate], they share this feature. Conversion yields resolution to masculine gender as shown in (57).

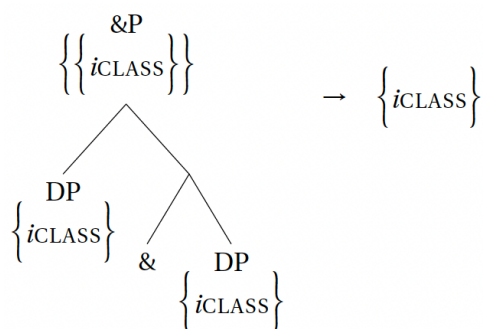
(57) Deriving masculine agreement in (56)a

Mismatch:



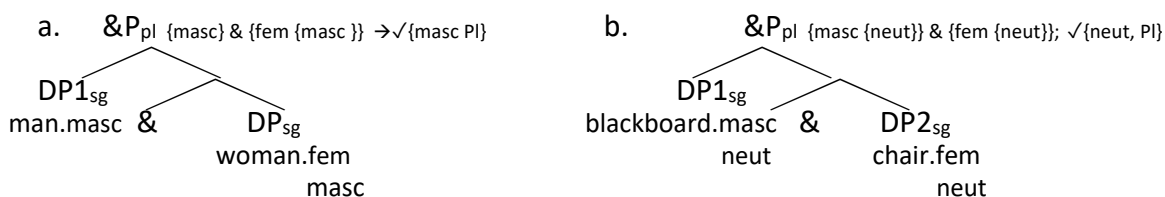
(56)b differs crucially in their account because the feminine and masculine gender features of inanimates are both uninterpretable, so they are ignored. Conversion finds [class] as the shared feature, and this yields resolution to neuter gender.

(83) Conjoined inanimates of mismatched genders in Greek = (56)b



As we've seen above in 5.2.1, this approach to arbitrary genders cannot be extended successfully to Xhosa; it yields the wrong results. But a unified approach is possible if we extend to Greek my approach to Xhosa: Greek outer *n*Ps of feminine and masculine gender would stack above different *i*cores depending on animacy. in (56)a the core is a masculine *n* with the *i*[animate] semantics that Adamson & Anagnostopoulou propose, and in (56)b, a neuter *i*[inanimate] *n*:

(84) Extending the stacked *n*P approach to Greek



Conjunctions of gender-matched DPs, whether arbitrary or semantic, are correctly predicted to pair with matching plural agreement just as in Xhosa (and as their account predicts as well). We saw this in (59), reproduced below. The systematic exception is genders with no *i*flavors at all.

(59) I fusta ke i bluza ine vromikes. [Greek; Adamson & Anagnostopoulou 2024:39]
 the.F.SG skirt and the.F.SG t-shirt are dirty.F.PL
 'The skirt and the t-shirt are dirty.'

5.2.3 Loose ends

Interestingly, Adamson & Anagnostopoulou report two instances in which Greek semantic

gender prevails over arbitrary gender in ways that lack Xhosa parallels.

First, in a small number of cases Greek [human] nouns have arbitrary gender. This is systematically ignored in agreement with conjuncts. In (85) ‘genius’ refers to a male individual though it is formally feminine, matching the semantically and formally feminine ‘sister.’ Only resolved masculine agreement is possible (85) = their (68), adapted), as it would be for a formally [MASC+FEM] conjunction, based on (57).

(85) I megalofiia ke i adherfi tu ine {charmeni. /*charumenes}.
the.F.SG genius(F.SG) and the.F.SG sister(F.SG) his are happy.M.PL/*happy.F.PL
‘The genius and his sister are happy.’

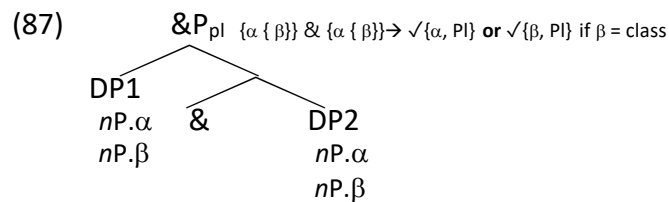
Adamson & Anagnostopoulou note that this does not fall out from their percolation mechanism, which allows either *i*Fs or *u*Fs (i.e. conceptual or arbitrary gender features) to percolate. Given this evidence that semantic agreement is obligatory for [human] nouns alone, they suggest that it holds of them as a consequence of Maximize Presupposition. There are no parallels to this pattern in Xhosa, where matching arbitrary gender features [α .sg + α .sg] of conjuncts pair exceptionlessly with matching [α .pl] agreement whatever the semantics of the conjuncts except for [3+3] and [5+5], as we have seen. As the Greek pattern does not fall out from my approach either, I leave this point of variation as a puzzle for future research.

Secondly, some Greek speakers accept neuter agreement with uniform masculine and uniform feminine conjunctions of inanimates. Thus alongside of (59), some accept (86):

(86) %I fusta ke i bluza ine vromika. [Greek; Adamson & Anagnostopoulou 2024:42]
the.F.SG skirt and the.F.SG t-shirt are dirty.NEUT.PL
‘The skirt and the t-shirt are dirty.’

Adamson & Anagnostopoulou propose that inanimates alone may percolate either *i*Fs or *u*Fs while animates must percolate *i*Fs.

In terms of the analysis I have presented, this suggests that for some Greek speakers, when DP1 and DP2 match in both their outer and their *icore* gender features, the higher does not necessarily win as I proposed for Xhosa in §5.1; there is some wiggle room available for variation with the result that agreement may reflect the semantic gender as illustrated schematically in (87). Since the intersection includes both α and β this can be addressed with a relatively minor stipulation restricting the option as needed (i.e. it is unavailable in Xhosa, and available only for inanimates in Greek).



6 Agreement with conjoined plurals: patterns and implications

6.1 Overview

In this section we will see evidence that resolution is never triggered by gender mismatch in Xhosa, contra Corbett & Mtenje (1987)'s proposal that this is generally the case: for the Xhosa speakers in my study, resolution does not apply to determine agreement with conjoined plurals at all. Not only does default agreement fail to appear in circumstances paralleling those where it is required with conjoined singulars; there is also variability in agreement with plurals whose genders mismatch to a degree not found with conjoined singulars. Both factors point towards a very different derivation without the resolution process.⁸

⁸ Corbett & Mtenje (1987) give a Chichewa example of class 8 *zi-* accompanying a conjunction of mismatched plurals from classes 6 and 4; *zi-* can be neither FCA nor LCA. Whether this is a cross-linguistic difference between Xhosa and Chichewa requires comparative study to determine. They also report that both gender-matching plural agreement and default *zi-* are acceptable in an example with [4+4] [inanimate] conjoined nouns in Chichewa. In other cases, *zi-* is ruled out or dispreferred. I leave it to future research to follow up.

Despite this, agreement largely avoids any plural conjunct of noun class 4 or 6 in favor of those drawn from class 2, 8, or 10. This parallels the inability of the singulars 3 and 5 to value agreement when conjoined, strongly supporting the analysis of 3/4 and 5/6 as connected -- as singulars and plurals of the same genders, contra Taraldsen et al (2018). The patterns that emerge are as follows, where (88)(i) bleeds (ii).

- (88) Hierarchies of preference in agreement with gender-mismatched conjoined plurals
 (i) Genders with *i*content > default > wholly *u*genders
 (ii) FCA > LCA

To capture these patterns I propose a Multiple Agree derivation in which the features of both conjuncts copy to the probe and are reduced to a single set for exponence, guided by these preferences. Multiple Agree is blocked if the conjuncts are singular because, as proposed by Marušič et al (2007) and Bošković (2009) and mentioned in §3.3, to interact with the conjuncts the probe must also target &P, and &P is plural. The conjuncts must be compatible with &P's plurality for Multiple Agree to succeed.⁹

6.2 When genders of plurals match

As noted in §5.1, all Xhosa gender-matching [plural+plural] conjunctions pair with expected grammatical agreement and not with defaults. We see this for classes 2, 8, and 10 in (89) where the pattern is not surprising, since singulars 1, 7, and 9 also pair with their expected plurals. But (90) and (91) show that the pattern holds even for [4+4] and [6+6]. Compare these to examples with conjoined singulars of the same genders, i.e. [3+3] and [5+5] in (81) and (82) above; the

⁹ Taraldsen et al (2018) mention a small number of speakers offering agreement with a singular conjunct in lieu of resolved agreement with mismatches (of approx. 20 speakers on each experiment, 1 speaker on one data set, 2 on another). Of the 8 speakers in my study none did so, but I have worked remotely since then with a speaker who often offers FCA with a singular especially in cases of ineffable mismatches. This suggests that &P is not always plural for all speakers; how this impacts the overall grammar of agreement with conjuncts remains to be explored.

conjoined singulars of these two genders pair with default *zi-* or *ba-*, depending on

humanness.¹⁰

- (89) a. Aba-ntwana noo-lolilwe **ba**-ya-hamba.
 2-children and.2a-trains 2SM-DISJ-move
 'The children and the trains are moving.'
- b. Izi-tyebi nezi-bhanxa **zi**-ya-funda ndawonye. [Mitchley 2015: 116, adapted]
 8-rich.persons and.8-fools 8SM-DISJ-study together
 'The rich people and the fools are studying together.'
- c. Iin-tombazana neen-kwenkwe **zi**-ya-sebenza. [9+9=10]
 10-girls and.10-boys 10SM-DISJ-work
 'The girl and the boys are working.'
- (90) a. Imi-pu nemi-bhobho **i**-se rumi-ni. [4+4=4]
 4-guns and.4-pipes 4SM-LOC room-LOC
 'The guns and the pipes are in the room.'
- b. Ama-sele nama-dada **a**-ya-qubha. [6+6=6]
 6-frogs and.6-ducks 6SM-DISJ-swim
 'The frogs and the ducks are swimming.'
- (91) a. Ama-polisa nama-gqwetha **a**-ya-sebenza ndawoye. [6+6=6]
 6-policemen and.6-lawyers 6SM-DISJ-work together
 'The policemen and the lawyers are working together.'
- b. Imi-gewu nemi-gulukudu **i**-se rumi-ni. [4+4=4]
 4-criminals and.4-gangsters 4SM-LOC room-LOC
 'The criminals and the gangsters are in the room.'

The examples inform us that the "problem" genders *can* factor into agreement with conjoined DPs just in case they belong to gender-matched plurals. The set of facts is surface compatible with Corbett & Mtenje's (1987) proposal that only singular number *or* mismatches in gender trigger resolution, since in conjunctions of [4+4] and [6+6], neither occur. But the next subsection presents reason to revise this, at least where Xhosa is concerned.

6.3 When genders of plurals mismatch

6.3.1 FCA with 2, 8, 10

In agreement with gender-mismatched [pl+pl] combinations, first conjunct agreement (FCA)

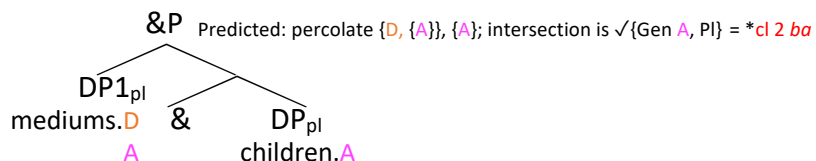
¹⁰ This difference between singulars and plurals seems logically reducible to the fact that singular is additive, while plural is cumulative. Combining singulars returns something different -- a plural -- but combining of plurals yields plurals; a calculation or transformation is involved in the first instance, but not in the second. I leave it to future research to explore this connection.

was preferred by a majority of the speakers in my study. (92) shows that this preference was absolute when the first conjunct was a human-denoting plural of class 2. When DP1 was instead class 8 or 10, the preference was not absolute but nonetheless strong. Even when the second conjunct was class 2 as in (93)a,b, five and six out of eight speakers selected FCA.¹¹

- (92) Aba-ntwana neen-tombi ba-ya-cula. [2+10: FCA chosen by 8 out of 8 speakers]
 2-children and.10-girls 2SM-DISJ-sing-FV
 'The children and the girls are singing.'
- (93) a. lin-tombi naba-ntwana zi/ba-ya-cul-a. [10+2: FCA>LCA 5:3]
 10-girls and.2-children 10SM/2SM-DISJ-sing-FV
 'The young ladies and the children are singing.'
- b. Iza-nuse naba-ntwana zi-ya-cul-a. [FCA>LCA 6:2]
 8-mediums and.2-children 8SM-DISJ-sing-FV
 'The young ladies and the children are singing.'
- c. Izi-tya nemi-nqathe zi-se tafile-ni. [FCA>LCA 6:2]
 8-plates and.4-carrots 8SM-be 9-table-LOC
 'The plates and the carrots are on the table.'

The results are not compatible with the conversion and resolution processes detailed in §5, which would yield class 2 *ba-* agreement in (93)a,b as (94) illustrates.

- (94) Prediction for (93)b is default *ba*; 6 out of 8 speakers choose FCA *zi-*



Underscoring the difference between these results and those for conjoined singulars, default agreement matching neither conjunct was strongly dispreferred. Table 2 shows the results for 12 pairs of [+human] plurals of which neither is drawn from class 2. Across eight speakers, class 2 *ba-* was chosen only 10 times out of 96 examples in which it matches neither

¹¹ In my margin annotations, [X>Y N:M] means conjunct X was favored over conjunct Y in that N chose X and M chose Y, where N and M are numbers out of 8 speakers in my study.

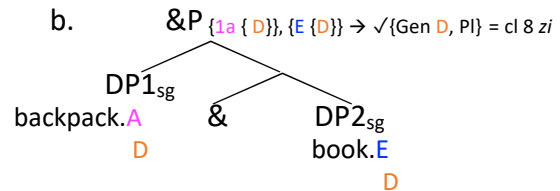
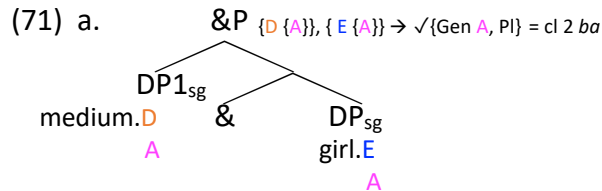
conjunct and thus is unambiguously default.¹² The upshot is that when one plural conjunct is class 2, *ba*-agreement is more than likely formal agreement with that DP.

Table 3: Default agreement for combinations in which no conjunct = class 2 (eight speakers)

4+6	4+8	4+10	6+4	6+8	6+10	8+4	8+6	8+10	10+4	10+6	10+8
2	2	2	2	0	0	1	0	0	1	0	0

Recall the pattern of default agreement when singulars are conjoined in cases like (70)a,b (reproduced below) for which I proposed the derivations in (71)a,b. In many cases including (70)a, agreement matches neither conjunct.

- (70) a. Is-anuse nen-tombi **ba**-ya-sebenza. b. U-bhaka nen-cwadi **zi**-ngaphandle.
 7-medium and.9-girl 2SM-DISJ-work 1a-backpack and.9-book 8SM-be.outside
 'The medium and the girl are working.'



I conclude that gender mismatch does not trigger resolution, contra Corbett & Mtenje (1987)'s proposal; at least, not in Xhosa. Only singular conjuncts do so.

6.3.2 LCA avoiding 4 and 6

The following examples show that if DP1 is class 4 or 6 and DP2 mismatches it, the preference shifts strikingly to last conjunct agreement (LCA). Agreement with first conjuncts of classes 4 and 6 is not ruled out, but it is clearly dispreferred in these examples. When there is a class 4 conjunct, default *zi*-or *ba*- often makes an appearance as these examples show.

- (95) Imi-gewu na-**ba**-ntwana **ba**/i/zi-ya-cul-a. [LCA > FCA & [-human] default 5:2:1]
 4-criminals and.2-children 2SM/4SM/8SM-DISJ-sing-FV
 'The criminals and the children are singing.'

¹² Interestingly in all of the cases where default *ba*- was chosen, one conjunct was class 4, which seems the "weakest" gender. I have no insights to offer but see §6.5 on a potentially relevant idiosyncrasy of class 6.

- (96) Ama-polisa na-**ba-ntwana** **ba**/i-ya-cul-a. [LCA > FCA 6:2]
 6-polisa and.2-children 2SM/6SM-DISJ-sing-FV
 'The policemen and the children are singing.'
- (97) Imi-gewu **neen**-tombi i/zi-/ba-ya-cul-a. [LCA & default > FCA 3:3:2]
 4-criminals and.10-girls 10SM/4SM-/2SM-DISJ-sing-FV
 'The criminals and the children are singing.'
- (98) a. Imi-ngathe **nezi-tya** zi-/i-se tafile-ni. [LCA > FCA 6:2]
 4-carrots and.8-plates 8SM/4SM-be 9-table-LOC
 'The carrots and the plate disappeared.'
- b. Ama-qanda **nezi-tya** a/zi-nyamalele. [LCA > FCA 7:1]
 6-eggs and.8-plates 6SM/8SM-disappeared
 'The eggs and the plate disappeared.'
- c. Imi-funo **neem-botyi** zi/i-phel-ile. [LCA > FCA 6:2]
 4-vegetables and.10-beans 10SM/4SM-be.finished-DISJ
 'The vegetables and the beans are finished.'

The greatest variability arises in conjunctions [4+6] and [6+4]. The plurality of my speakers chose default *zi*- and *ba*- depending on animacy; otherwise, FCA slightly edged out LCA:

- (99) Imi-ngathe **nama-qanda** i-/a-/zi-se tafile-ni. [default *zi* > FCA > LCA 5:2:1]
 4-carrots and.6-eggs 4SM/6SM/8SM-be 9-table-LOC
 'The carrots and the eggs are on the table.' [i.e. *zi* x 5, *i* x 2, *a* x 1]
- (100) Ama-qanda **nemi-ngathe** i-/a-/zi-se tafile-ni. [default *zi* > FCA or LCA 3:2:2]
 6-eggs and.4-carrots 4SM/6SM/8SM-be 9-table-LOC
 'The eggs and the carrots are on the table.' [i.e. *zi* x 3, *i* x 2, *a* x 2]
- (101) Imi-gewu **nama-polisa** i-/a-/ba-sebenza ndawonye. [default *ba* > LCA > FCA 4:3:2]
 4-criminals and.6-police 4SM/6SM/2SM-work together
 'The criminals and the police are working together.' [i.e. *ba* x 4, *a* x 3, *i* x 2]

6.4 Analysis of agreement with conjoined plurals

I summarize the patterns to be captured in (102).

- (102) a. Agr with gender-matching plurals is gender- and number-matching
 b. Agr with mismatched plurals is preferentially FCA *but*
 c. Iff DP1 bears a wholly *u*gender, the preference changes to LCA *and*
 d. Iff DP1 & DP2 both bear wholly *u*gender, results are highly variable. Default agreement is an option in this case and is the plurality choice.

That a conjunction of plurals [4+4] or [6+6] yields matching *i*- or *a*- agreement and not default *ba*- or *zi*- suggests that percolation and resolution are last resort in Xhosa. Adapting

Marušič et al (2007) and Bošković (2009) I propose that the preferred mechanism for

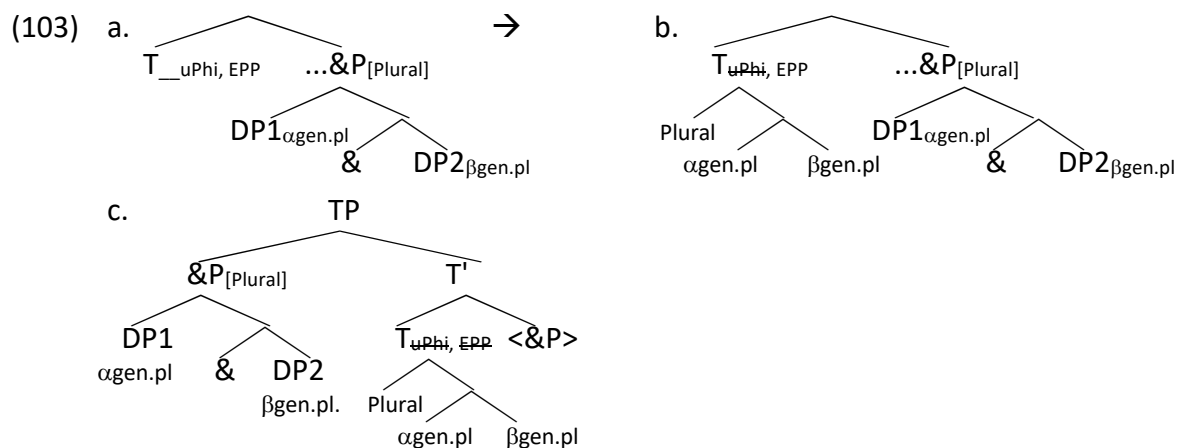
Let us see why we need Multiple Agree to copy the features of both conjuncts, and how it derives the results in three contrasting cases presented in the subsections above and reproduced below. The favored pattern in (93)b and (90)a could be derived by Multiple Agree including only the first conjunct – Agree (T, &P, DP1) -- as DP1 suffices to provide the features realized in subject agreement here. But that would erroneously yield class 4 agreement in (98)c as well, where in actuality the preference is for LCA. A minimal modification could accommodate this example: if Agree (&P, DP1) applies and encounters a wholly *ugender* in DP1, probing might move on to DP2 only in that case (perhaps deleting the *ugender* of DP1). This will yield FCA in (93)b and LCA in (98)c. In (90)a, where DP2 also has *ugender* (the same one as DP1), we might suppose that as the last available option DP2's feature is preferentially expounded.

- ¹³ Three points: (i) Following Bosković (2009), the fact that conjuncts must be plural under Multiple Agree is evidence that the relevant uPhi probe is not a “split” probe that can obtain a number value from one goal and a gender value from another. (ii) The obligatoriness of agreement is likely due to labeling requirements (Chomsky 2013, 2015). I leave follow-up to future research. (iii) A prediction of the last resort approach to resolution, to explore in future work: if there is variation in whether &P is always plural, there should be accompanying variation in whether resolution is employed – a language might sometimes or never utilize it.

But then, (99) and (100) present a puzzle: the procedure wrongly predicts preferential agreement with DP2 since DP1 has *u*gender, but instead, this is the one circumstance in which default agreement predominates.

- (99) Imi-ngathe **nama-qanda** i-/a-/zi-se tafile-ni. [default *zi*>FCA>LCA 5:2:1]
 4-carrots and.6-eggs 4SM/6SM/8SM-be 9-table-LOC [i.e. *zi* x 5, *i* x 2, *a* x 1]
 'The carrots and the eggs are on the table.' [4+6=default 8]
- (100) Ama-qanda **nemi-ngathe** i-/a-/zi-se tafile-ni. [default *zi*>FCA or LCA 3:2:2]
 6-eggs and.4-carrots 4SM/6SM/8SM-be 9-table-LOC [i.e. *zi* x 3, *i* x 2, *a* x 2]
 'The eggs and the carrots are on the table.'

I conclude from this pattern that the features of both conjuncts factor into the choice of agreement simultaneously. Both sets are copied to the probe, establishing a reproduction of the features of the two conjuncts.¹⁴ The gender features' relative left-to-right positions in the *u*Phi bundles below reflects the hierarchical or linear relations among conjuncts.



Two instances of a single feature are simply collapsed to one set (adapting Adamson & Anagnostopoulou on conversion of identical features), yielding matching agreement in (90)a.

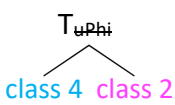
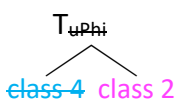
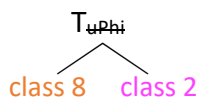
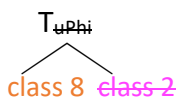
When gender features mismatch, because T can expone only a single gender feature, an

¹⁴ See also Citko (2018) for an approach copying features of both conjuncts onto the probe and Akkus (2020) for a Multiple Agree analysis of a phi-feature dependency between subject and object in Iranian.

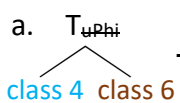
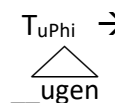
impoverishment operation (Bonet 1991) must delete one of them.^{15,16} The hierarchies in (88) determine which gender feature is preferentially retained, with (i) bleeding (ii). I assume these are language-specific; FCA is not a consistent cross-Bantu preference (Mitchley 2015).

- (88) Hierarchies of preference in agreement with preverbal conjoined plurals
 (i) Genders with *i*content > default > wholly *u*genders
 (ii) FCA > LCA

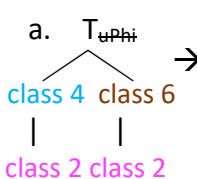
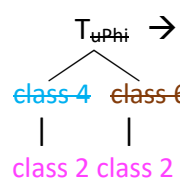
Application of the impoverishment operation is sketched in in (104).

- (104) a.  $\rightarrow$  by HoP (i) *i* > *u*-content
 b.  $\rightarrow$  by HoP (ii) FCA > LCA

Turning to default agreement with conjoined plurals like (99)-(101), I suggest that there is deletion of both gender values on the probe. This might simply render its *u*gender unvalued and force default insertion to take place:

- (105) a.  $\rightarrow$ b.  $\rightarrow$ c.  $\rightarrow$ d.  $\rightarrow$ default class 2 or 8

Alternatively, it might be that when the features of each conjunct copy to the probe, their core layers are included; deletion of the higher genders exposes them. This would yield (106) as the derivation for default *ba*-agreement in a case like (101), reproduced below. I leave this choice open.

- (106) a.  $\rightarrow$ b.  $\rightarrow$ c. 

¹⁵ Murphy & Puškar (2018) also apply impoverishment to derive the features of agreement with conjuncts. In their account, impoverishment reduces multiple gender values on &P itself, to arrive at the least marked.

¹⁶ For the speakers that I consulted, conjunctions of more than two DPs always pair with default/semantic agreement. This might indicate a limitation of the Multiple Agree process; I leave it to future research.

(101) lmi-gewu nama-polisa i-/a-/ba-sebenza ndawonye.
 4-criminals and.6-police 4SM/6SM/2SM-work together
 'The criminals and the police are working together.'

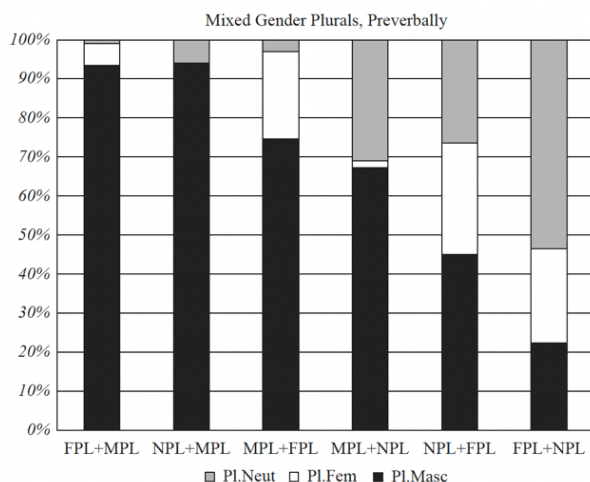
[default ba > LCA > FCA 4:3:2]
 [i.e. ba x 4, a x 3, i x 2]

Summing up, agreement with conjoined plurals exhibits patterns different from those of conjoined singulars. The strategies I have described in this section do not work for conjoined singulars since Multiple Agree fails for them as we have seen; only percolation and resolution can value the probe in such cases.

6.5 Conjoined plurals in existing approaches

It is already well-established that gender agreement with conjoined plurals varies across speakers. Table 3 represents responses on different combinations in Slovenian.

Table 3 Slovenian conjoined plurals (Marušić et al 2015)



There are a number of approaches to the variability of judgments on conjoined plurals. Marušić et al (2015), Marušić & Nevins (2020) propose a Distributed Agree model under which agreement applies in two stages, Agree Link and Agree Copy. If Agree Copy applies before linearization, the result is highest conjunct agreement; if Agree Copy applies after linearization then there is closest conjunct agreement because the structure is flattened out. I represent this schematically in (107) and (108).

(107) Agree-Copy before linearization: default gender or agreement with DP1, regardless of word order, because what's visible to syntax is hierarchical structure



(108) Agree-Copy after linearization: default gender or closest conjunct agreement, because hierarchical structure ceases to be visible



This approach predicts random results, but as we have seen, agreement with conjoined plurals in Xhosa is variable but not random. To capture the Xhosa hierarchies of preference for conjoined plurals would require massive look-ahead establishing, for instance, that DP2 is more agreeable than DP1, making it preferable to wait and do Agree Copy after linearizing.

A similar problem arises in Murphy & Puškar (2018), who propose that the head & can in principle obtain multiple gender values from its conjuncts by Agreeing with them. Which values it acquires depends on the order of application among the operations Merge, Agree Up, and Agree Down. If Agree Up applies before Merge, and then Agree Down applies after Merge, there will be LCA because Agree Up applies vacuously before the first conjunct is present. Agree Down will successfully give &P the gender feature of its second (lower) conjunct, so the result must be LCA. Agree Down preceding Merge with Agree Up applying last yields FCA, because in this case Agree Down is vacuous, but Agree Up is successful. Hence & has features of the first conjunct and agreement must be FCA.

Like Distributed Agree, this provides no handle on the way the gender features of the conjuncts impact the choice of FCA vs LCA in Xhosa. And in fact, the Slovenian results depicted

in Table 4 are also variable without being random. When there is a masculine conjunct, masculine agreement prevails. And for any given gender, the odds of its determining agreement are better if it is borne by the second conjunct than by the first. Neuter seems more likely to be adopted than either feminine or the masculine default, when neuter and feminine are combined. Feminine is more likely to be chosen when it's the last conjunct than when it is the first, but it is the least likely overall. Something along the lines of (109) seems to be the pattern:

- (109) Draft hierarchy of preferences for agreement with conjoined plurals in Slovenian
 (i) masc > neuter > default > feminine
 (ii) LCA > FCA

It lies outside the scope of this brief section to delve further into these preferences, their basis, and how they interact. The point I wish to make is that trends are detectible for Slovenian in Table 4 as well as in the Xhosa data that we've explored. These cannot be captured in an approach designed to generate randomly variable results.

6.6 For future research

Before leaving conjoined plurals I will describe a loose end that I must regretfully leave unaccounted for, having to do with [human] nouns of plural class 6.

When [human] nouns are conjoined, DP1 is class 6 and DP2 is of class 8 or 10, FCA predominates as shown in (110)a,b. this is unlike what happens in the case of an [inanimate] class 6 first conjunct, which shifts the preference to LCA under (88)(i), the same as does a class 4 first conjunct. The pattern for a [human] DP1 of class 6 is more mixed than when classes 2, 8, or 10 are DP1, but nevertheless it favors FCA. The examples in (98) (repeated below) illustrate the contrasting pattern for [inanimate] DPs, where DP1 is class 6 or 4.

- (110) a. Ama-ggirha nez-anuse a-/ba/zi-sebenz-a ndawonye. [FCA>default>LCA]
 6-doctors and.8diviners 6SA/2SA/8SA-work-FV together
 'The traditional doctors and the diviners are working together.'

[results: FCA x 5, default *ba*- x 1, FCA or default x1, LCA x 1]

- b. Ama-polisa neen-tombi a-ya-cul-a. [FCA>LCA>default]
 6-police and.10-girls 6SA/ -DISJ-sing-FV
 'The policemen and the young ladies are singing.'
 [results: FCA x 4, LCA x 2, FCA or LCA x 1, default *ba*- x 1]

- (98) a. Imi-ngathe nezi-tya zi-i-se tafile-ni. [LCA >FCA: 6:2]
 4-carrots and.8-plates 8SM/4SM-be 9-table-LOC
 'The carrots and the plate disappeared.'
- b. Ama-ganda nezi-tya a/zi-nyamalele. [LCA >FCA 7:1]
 6-eggs and.8-plates 6SM/8SM-disappeared
 'The eggs and the plate disappeared.'
- c. Imi-funo neem-botyi zi/i-phel-ile. [LCA>FCA: 6:2]
 4-vegetables and.10-beans 10SM/4SM-be.finished-DISJ
 'The vegetables and the beans are finished.'

In classes other than class 6 -- 1/2, 3/4, 5, 7/8, 9/10 – [human], [animal], and [inanimate] nouns pattern alike in crucial respects: if gender-matched singulars are conjoined, agreement is gender-matching plural regardless of the animacy of the singulars; all [4+N] pair with LCA regardless of the animacy of the class 4 first conjunct. What can be the source of the exceptional behavior of class 6 in (110)a,b? There is a potential clue in the fact that class 6 serves as plural for many [human] nouns whose singulars fall in various classes.¹⁷

- | | | |
|---|---|---|
| (111) a. in-doda/ama-doda
9-man/6-man
'man/men' | b. in-kosi/ama-khosi
9-king/6-king
'king/kings' | c. in-kosilazi/ama-kosilazi
9-wife/6-wives
'wife/wives' |
| d. um-Xhosa/ama-Xhosa
1-Xhosa/6-Xhosa
'Xhosa person/Xhosa people or nation' | e. um-Zulu/ama-Zulu
1-Zulu/6-Zulu
'Zulu person/Zulu people or nation' | |

The translations in (111)d and (111)e reflect a kind of 'group' reading often associated with class 6 plurals of [human] nouns. I propose that these plurals correspond to a distinct gender F which has the meaning 'group' underlying a prefix and agreement class that I'll call 6x, homophonous with the class 6 plural of gender C but combining only with a marked subset of

¹⁷ In some Bantu languages class 6 is a generic plural, combining with nouns of many semantic properties and classes; the Xhosa pattern is more restrictive.

[human] nouns.

(112) ns for Xhosa, reflecting conjunct agreement facts

Classes 1/2 = Gender A	n_{A1}	i [human]
	n_{A2}	arb
Classes 3/4 = Gender B	n_B	uninterpretable, for all members
Classes 5/6 = Gender C	n_C	uninterpretable, for all members
Classes 7/8 = Gender D	n_{D1}	i [artifact]
	n_{D2}	arb
Classes 9/10 = Gender E	n_{E1}	i [animal]
	n_{E2}	arb
Class 6x = Gender F	n_F	i [human group]

Because it is interpretable, gender F/6x falls under preference (i) of (88). But [human] nouns pluralized with *ama-* are ambiguous, leaving the door open to some variability.

This leaves (96) to be explained: why is FCA not the choice of four or five speakers, as it is in (110) where DP2 is class 8 or 10? Instead, LCA is favored.

- (96) Ama-polisa na-**ba-ntwana** **ba**/i-ya-cul-a. [LCA > FCA 6:2]
6-polisa and.2-children 2SM/6SM-DISJ-sing-FV
'The policemen and the children are singing.'

It seems that class 2, a canonically [human] interpretable gender, outranks the ambiguously (un)interpretable [+/-human] class 6(x). This accords with the prominence of [human] in feature hierarchies (Hammerly 2023 among others). But I leave it to future research to find an insightful way to integrate it into the account of agreement with conjoined plurals.

7 Agreement failure and semantic exceptionality

7.1 Agreement and special semantics

This section considers the question of what precisely “uninterpretable” means in relation to classes 3/4 and 5/6: do they really lack interpretable semantic content, or rather, drawing on Denny & Creider (1986) among others, do they have content incompatible with the grammar of agreement with conjoined singulars? Building on Arsenijević (2017, 2021), Adamson & Anagnostopoulou (2024) relate the failure of BCS *[neut+neut= neut.pl] to the fact that many

neuters are mass nouns, lacking individuation and number. Neuters pluralize irregularly or not at all; neuter is incompatible with certain deictic functions; and pronominal reference and pronominal binding with neuter nouns exhibit departures from the regular patterns found in other genders. I will exemplify some of these empirical factors in discussion below. It is worth exploring whether Xhosa classes 3/4 and 5/6 might exhibit a similar range of irregularities given Denny & Creider's proposed meanings of solid figure/(non-)extended shape for these pairs.

Supporting evidence turns out to be elusive. Classes 3 and 5 are not among those lacking plurals or devoted to abstracts; while some nouns of these classes conform to Denny & Creider's proposed shape attributes, they are mostly ordinary objects that are countable and that pluralize regularly as shown in (113).

- (113) a. **um**-thi **omu**-nye b. **imi**-thi **imi**-thathu c. **i**-qanda **eli**-nye d. **ama**-qanda **ama**-thathu
 3-tree 3-one 4-trees 4-three 5-egg 5-one 6-eggs 6-three
 'one tree' 'three trees' 'one egg' 'three eggs'

Apart from class 1, every Xhosa class includes some nouns referring to uncountable substances such as liquids and/or masses as I showed in (17), repeated below.

(17) Liquids and masses: scattered

- | | | | | | |
|---|--|--|---|-------------------------------------|--|
| a. um -chamo
3-urine
'urine' | b. i -gazi
5-blood
'blood' | c. ama -nzi
6-water
'water' | d. isi -dudu
7-porridge
'porridge' | e. i -oili
9-oil
'oil' | f. u -bisi
11-milk
'milk' |
| g. um -gubo
3-flour
'flour' | h. ama -zimba
6-sorghum
'sorghum' | i. i -tyuwa
9-salt
'salt' | j. u -daka
11-mud
'mud' | | |

These nouns must combine with a unit counter for enumeration to be possible as shown in (114)-(115). But (114)c shows that the problem arises for a liquid in class 9 – a class in which conjoined singulars take regular plural agreement. Moreover, my consultants offered as the counter for liquids *ithonzi* - 'drop' in (115), itself a noun of the problematical class 5. Taken together this and the examples in (113) argue strongly against any deficiency in countability

correlating with the need for default agreement with conjoined singulars of classes 3 and 5. If countability is foundational to successful agreement, it does not seem to be lacking here.

- (114) a. in-cwadi e-nye b. in-tombazana e-nye c. #i-oili e-nye
 9-book 9-one 9-girl 9-one 9-oil 9-one
 'one book' 'one girl' Literally: 'one oil'
- (115) a. #i-gazi eli-nye b. i-thonzi (eli-nye) l-e-gazi / l-e-oili
 5-blood 5-one 5-drop 5-one 5-of.5-blood. /5-of.9-oil
 Literally: 'one blood' 'a/one drop of blood/oil' [unit counter is cl.5]

Arsenijević (2017, 2021) provides an additional diagnostic for defectiveness of BCS nouns in neuter gender: he reports that neuter is unavailable in “highly individuated plural uses such as....in demonstratives with pointing...” ((116) from Arsenijević 2021:21). (117)a-d show that this is not a problem for classes 5/6 and 3/4, whether singular or plural. If individuation is also crucial to successful agreement with conjoined singulars, there is no absence in these classes that could account for its failure.

- (116) Ona *(deca) su neumorna. (pointing at children)
 those.NEUT.PL child. NEUT.PL AUX.PL tireless.NEUT.PL
 'Those children are tireless'/*'Those are tireless.'
- (117) a. Loo (m-thi /m-gulukudu) m-de. b. Eli (qanda/gqwetha) l-awa.
 3that 3-tree/3-gangster 3-tall 5that 5egg/5lawyer 5SM-fell
 'That (tree/gangster) is tall' 'That (egg/lawyer) fell'
- c. Le (mi-thi /mi-gulukudu) im-bi. d. La (ma-qanda/ma-nenekazi) ma-hle.
 4those(4-trees/4-gangsters) 4-bad 6those (6-eggs/6-ladies) 6-nice
 'Those (trees/gangsters) are bad.' 'Those (eggs/ladies) are nice.'

Pronominal reference and pronoun binding are also relevant tests. In certain languages including German, where a noun’s grammatical neuter gender is at odds with its semantic gender, a pronoun may match the latter rather than the former. Example (118) is from Mann 1979 via Braun & Haig 2010 (glosses added). The phenomenon is compatible with the idea that neuter is morpho-syntactically special, or perhaps semantically impoverished.

- (118) Das junge Mädchen gab ihm die Hand. Sie wollte wohl nett sein, [German]
 the young girl_{neut} shook him the hand. she_{fem} wanted probably nice be

aber was war mit ihr anzufangen.
but what was with her_{fem} start

'The young girl shook his hand. Apparently, she intended to be nice, but what could be done with her.'

In Xhosa this pattern is not replicated -- in fact, formal noun class matching is very strict; I am told that non-matches yield disjoint readings. (119) is a discourse in which subject and object markers match a class 5 human-denoting noun, and (120) is a parallel discourse involving class 10. In neither case can a pronoun or agreement marker of canonically human classes 1/2 be felicitously substituted.

- (119) I-gwetha I-asondela kwi-jury. L- /#u-ashwankathela i-tyala.
5-lawyer 5SM-approach LOC.9-jury 5SM- /#1SM -summarized 5-case
'The lawyer approached the jury. He summarized the case.'

Ba-li-/#m-phulaphula ngenyameko.
2SM-5OM-/#1OM listen carefully
They listened carefully to him.'

- (120) lin-tombi z-a-thenga ama-hashe. Z-a-zi-nga-kwazi uku-wa-khwela.
10-girls 10SM-buy 6-horses 10SM-PST-10SM-NEG-know 15-6OM-ride
Aba-hlobo ba-zo /#bo ba- be- zi- /#ba-hleka.
2-friends 2-10POSS-/2POSS 2SM-PST-10OM/#2OM-laugh
'The girls bought horses. They didn't know how to ride them.
Their friends laughed at them.'

Bound readings for Xhosa pronouns also require noun class matching, in every class; in contrast, a semantically feminine but formally neuter noun in Greek can bind either a neuter or feminine pronoun (see (122) from Spathas 2010:222).¹⁸

- (121) a. I-nenekazi nga-li-nye_i I-a-yi-funda in-cwadi ya-lo_i /ya-khe_j. *i = j
5-lady each-5-one 5SM-PST-9OM 9-book 9-5POSS /9-1POSS
'Each lady_i read her_i book.'
- b. Y-onke in-kwenkwe_i y-a-tya imi-funo ya-yo_i /ya-khe_j. *i = j
9-every 9-boy 9SM-PST-eat 4-vegetables 4-9POSS /4-1POSS
'Every boy ate his vegetables.'

¹⁸ In most cases gender-matching seems the norm between a pronoun and its binder even where the gender is arbitrary (see the French (i)); it is a formal requirement and not related to gender (flavor) interpretability.

(i) Chaque voiture a perdue sa /*son roue
every car.f aux lost 3s.f/*3s.masc wheel

c. Y-onke imi-gulukudu; y-a-khwela ama-hashe a-yo_i /a-bo_j. *i = j
 4-all 4-gangster 4SM-PST-ride 6-horses 6-4POSS /6-2POSS
 'All the gangsters rode their horses.'

(122) Kathe koritsi diakosmise to dhomatio {tis/tu} [Greek]
 every girl.N decorated the room hers/its
 'Every girl decorated her room.'

Summing up, Xhosa genders 3/4 and 5/6 display no symptoms of irregularity or defectiveness apart from the failure of matching agreement with [3+3], [5+5] which it is the goal to derive. That semantic features associated with a given gender interfere in resolution is unfalsifiable conjecture if there are no diagnostics confirming it. The properties of BCS neuter identified as telltale by Arsenijević (2017, 2021), Adamson & Anagnostopoulou (2024) have no parallels in the Xhosa genders whose singulars cannot pair with matching plural agreement, so it is not even clear whether those properties are causal or coincidental in the case of BCS neuter. Given that Xhosa genders 3/4 and 5/6 are ignored in the resolution process, I conclude that they are uninterpretable. I do not intend to claim that these genders have literally no semantic associations, but rather that they have none that the morphosyntax recognizes.

7.2 Evidence from Shona

In Xhosa we've seen that gender-matching conjoined singulars pair with matching plural agreement in three out of five of the genders considered; [3+3] and [5+5] are the sole exceptions. In BCS, singulars from two out of three genders pair with expected agreement; neuter is the sole odd man out. These patterns give the impression that matching agreement is the norm and its failure the exception to be explained.

But the Shona language raises a question regarding the generality of this. In Shona, the genders pairing with default agreement outnumber those in which matching agreement succeeds. The facts suggest that genders which *do* control matching agreement are the

exceptional ones, because of their relation to a very small number of semantic primitives. The large number of genders in Shona permits this alternative insight.

Shona includes eight regular singular/plural pairs of noun classes – three more than Xhosa, plus a few singletons. Conjunctions [1+1] and [7+7] pair with matching agreement in Shona as they do in Xhosa:

- (123) Mu-rume ne mu-kadzi **va-ri** panze. [Shona: 1+1=2]
 1-man and 1-woman 2SM-be outside
 ‘The man and the woman are outside.’
- (124) Chi-ngwa ne chi-bage **zvi-ri** pa-tafura. [Shona: 7+7=8]
 7-bread and 7-maize 8SM-be LOC-table [inanimate]
 ‘The bread and the maize are on the table.’
- (125) Chi-dhakwa nechi-komana **zvi-ri** ku-netsana. [Shona: 7+7=8]
 7-drunkard and.7-small.boy 8SM-be 15argue [human]
 ‘The drunkard and the small boy argued.’

[Human] and [inanimate] nouns of [3+3] and [5+5] pair only with agreement of classes 2 and 8 respectively, just as in Xhosa:

- (126) Mu-goti ne mu-ti Xi /√**zvi-ri** panze. [Shona: 3+3=8]
 3-cooking.stick and 3-tree 4SM/ 8SM-be outside
 ‘The cooking stick and the tree are outside.’
- (127) Dombo ne zai Xa /√**zvi-ri** panze. [Shona: 5+5=8]
 5stone and 5egg 6SM/ 8SM-be outside
 ‘The stone and the egg are outside.’
- (128) Benzi ne dinga Xa- /√**va-ri** ku-shayikwa. [Shona: 5+5=2]
 5fool and 5dimwit 6SM/ 2SM-be 15-missing
 ‘The fool and the dimwit are missing.’

But [9+9] nouns do not pair with class 10 agreement like they do in Xhosa. This is easily determined because class 8 and 10 agreements are distinct in Shona, *zvi-* vs. *dzi-*, unlike in Xhosa where both are *zvi-*.¹⁹ In Shona, class 9 nouns show the pattern illustrated for classes 3 and 5 in (126)-(128): plurals take class 10 agreement (see (129)a), but [9+9] conjunctions do

¹⁹ It is not the case that syncretism in Xhosa camouflages a similar pattern, since recall that [human], [animal], and [inanimate] nouns of Xhosa class 9 all combine with *zi-* when conjoined, just as all [1+1] pair with 2agr and [7+7] pair with 8agr.

not. Conjoined [human] nouns take class 2 *va*-agreement as shown in (129)b, and the rest – animals, body parts, artifacts and abstracts – pair with class 8 when conjoined ((129)c-f).

- (129) a. Im-bwa **dz**-aka-on-ana. [Shona: 10 = *zi*-]
 9-dog 9_{SM}-PST-fall
 'The dogs saw each other'
- b. N-herera ne n-yanzvi **va**-/***dz**-aka-on-ana. [Shona: 9+9=2]
 9-orphan and 9-expert 2_{SM}-/*10_{SM}-PST-see-RECIP
 'The orphan and the expert saw each other.'
- c. Im-bwa nhema nem-hou chena **zv**- /***dz**-aka-rwa. [Shona: 9+9=8]
 9-dog 9-black and.9-cow 9-white 8_{SM}-/*10_{SM}-PST-fall
 'The black dog and the white cow fell.'
- d. M-huno iyi ne nzeve iyi **zvi**- /***dzi**-ne utachiona. [Shona: 9+9=8]
 9-nose 9-this and 9-ear 9-this 8_{SM}-/*10_{SM}-have 14-infection
 'This nose and this ear are infected.'
- e. M-biya iyi ne sando iyo **zvi**- /***dzi**-ne n-gura. [Shona: 9+9=8]
 9-bowl 9-this and 9-hammer 9-that 8_{SM}-/*10_{SM}-have 9-rust
 'This bowl and that hammer are rusty.'
- f. N-yota ne n-zara **zv**- /***dz**-aka-tambudza va-fambi. [Shona: 9+9=8]
 9-hunger and 9-thirst 8_{SM}-/*10_{SM}-afflict 2-travelers
 'Hunger and thirst afflicted the travelers.'

Shona has class 11 nouns which pluralize in class 10. Like [9+9], conjoined [11+11] nouns combine with default class 8 *zvi*- and not with the class 10 *dzi*- which agrees with their plurals:

- (130) Ru-oko noru-rimi **zvi**-/***dzi**-ne u-tachiona. [Shona: 11+11=8]
 11-hand and.11-tongue 8_{SM}-/*10_{SM}-have 14-infection
 'The hand and the tongue are infected.'

Nouns of class 14 pluralize in class 6, but conjoined class 14 singulars pair with class 8 *zvi*-, not with the class 6 *a*- subject marker that goes with their plurals (see (131)a,b):

- (131) a. U-ta ne u-tanho **zvi**-ri/***a**-ri kunze. [Shona: 14+14=8]
 14-bow and 14-ladder 8_{SM}-/*6_{SM}-be outside
 'The bow and the ladder are outside.'
- b. Ma-uta ne ma-utanho **a**-ri kunze.
 6-bows and 6-ladders 6_{SM}-be outside
 'The bows and the ladders are outside.'

Diminutives are formed with the prefixes *ka*-/*tw*a- of classes 12/13. Their plurals pair with *tw*a- agreement (see (132)a,b). But (133) shows that when diminutives conjoin, agreement

must be taken from class 2 or 8.²⁰

- (132) a. mu-sikana → ka-sikana b. Twa-sikana **twa**-nyangadika.
 1girl 12-girl 13-girls 13SM-disappear
 'girl' 'tiny girl' 'The tiny girls disappeared.'
- (133) a. Ka-sikana ne ka-kómáná **va**-/***twa**-nyangadika. [Shona: 12+12=2]
 12-girl & 12-boy 2SM/*12-disappear
 'The tiny girl and the tiny boy disappeared.'
- b. Ka-mba ne ka-motokari **zva**-/***twa**-nyangadika. [Shona: 12+12=8]
 12-house & 12-car 8SM/*13SM-disappear
 'The tiny house and the tiny car disappeared.'

In contrast, Shona conjunctions of gender-matched plurals pair with matching plural agreement as we saw to be true of plural co-ordinations in Xhosa.²¹

- (134) a. Twa-sikana ne twa-komana **twa**-/***va**-nyangadika. [Shona: 13+13=13]
 13-girls and 13-boys 13 /*2SM-disappear
 'The tiny girls and the tiny boys disappeared.'
- b. M-huno idzi ne nzeve idzodzo **dzi**- /***zvi**-ne utachiona. [Shona: 10+10=10]
 10-nose 10these and 10ear 10those 10SM-/*8SM-have 14-infection
 'These noses and those ears are infected.'

For conjoined nouns of classes 11 and 14 one might be tempted to relate the agreement failures to the reduced number of dedicated plural classes in some way, since nouns of these classes pluralize in classes 10 and 6 respectively. But such an approach would not generalize to the failures of [9+9=10] and diminutive [12+12=3].

Taking [11/10] and [14/6] to be genders the same as other singular/plural pairings, Shona has nearly three times the number of genders as better-studied European languages, and several more than the five main singular/plural pairings found in Xhosa. But of these, only two show regular plural agreement with conjoined singulars. This suggests that it is the latter pattern that is the exceptional one, which would follow naturally from assuming that the

²⁰ There is also augmentative formation correlating with classes 21/6, but for these I presently have no data.

²¹ See Dechaine et al (2014) for arguments that Shona noun class (prefixes) are multi-functional, having a range of derivational functions in addition to sg/pl. This state of affairs is orthogonal but compatible with the view that only a small number of semantic oppositions are recognized in the morpho-syntax of agreement.

number of genders with semantic associations grammaticized in the relevant way is typically small. In light of this, it seems likely that even if Denny & Creider (1986) are shown to be incorrect in that tree and plant semantics are the best predictors for membership in classes 3/4, it would not matter. The morphosyntax recognizes only a very limited set of linkages between genders and meanings. Shona classes 9/10 include the plurality of animal terms as they do in Xhosa, but agreement with [9+9] is default, arguing that [animal] is not a relevant feature.

There is perhaps an analytical puzzle in that diminutive gender coincides strongly with a meaning. Diminutive 12/13 clearly is not one of the *icore* genders and, following Carstens (1991, 2008) I suggest that the diminutive meaning is contributed by a silent morpheme – a root, pairing with a gender that is uninterpretable as shown in (135) (see Creemers & Fenger 2018, Lowenstamm 2015, Simonović 2022, Wiltschko & Steriopolo 2007 on (some) derivational affixes as roots). This approach seems to align slightly better with the facts of agreement in (133) than is attribution of the diminutive meaning to a dedicated *n* of classes 12/13 (Kramer 2015, Fuchs & Van der Wal 2022). But either approach is accommodated by my proposal that only a few strands of meaning are intrinsic interpretable properties of genders, factoring into the morpho-syntax of gender agreement.

- (135) a. diminutive structure:
-
- $$\begin{array}{c}
 nP_{uG, iA} \\
 \swarrow \quad \searrow \\
 unG \quad \quad VP \\
 \quad \quad \swarrow \quad \searrow \\
 \quad \quad V_{DIMIN} \quad nP \\
 \quad \quad \quad \quad \swarrow \quad \searrow \\
 \quad \quad \quad \quad inA \quad VGIRL
 \end{array}$$
- b.
-
- $$\begin{array}{c}
 \&P_{pl, \checkmark, [Gen\ A, Pl]} \\
 \swarrow \quad \searrow \\
 DP1_{sg} \quad \quad VP \\
 \text{tiny girl}_{uG, iA} \quad \quad \swarrow \quad \searrow \\
 \quad \quad \quad \& \quad \quad DP2_{sg} \\
 \quad \quad \quad \quad \quad \text{tiny boy}_{uG, iA}
 \end{array}$$

Summing up, the Shona facts motivate the proposal I introduced in §4.1: that a small number of genders with recognized association to a culturally salient group of entities like [male], [female], [human], [animal], [artifact] may be special in having formal legibility in the syntax of

a language in a way that other genders don't. I have classified the former type of gender as having interpretable content in the syntax of the relevant language, and the others as uninterpretable. It seems clear that what is or is not adopted as interpretable varies across languages somewhat arbitrarily.

8 Conclusions

I have argued that a modest but grammatically significant semantic core underlies the Bantu noun class system: [human], [inanimate], and in Xhosa (but not Shona), [animal]. So-called default or semantic agreement with conjoined singulars in Bantu languages is actually syntactic agreement with one of these *igender* cores.

Singular nouns from only a small number of genders pair with gender-matching plural agreement when conjoined, and these are the genders of singular/plural pairings 1/2, 7/8, and in Xhosa, 9/10 – the ones with the above semantic associations. I have proposed that it is systematic uninterpretable of a gender that leads to failure of agreement with conjunctions of Xhosa [3+3] and [5+5] nouns and the several additional pairs that pattern like them in Shona; I propose to extend the approach to BCS neuter.

Given that there is uniformly matching plural agreement with any conjoined matched singulars drawn from Xhosa classes 1, 7, and 9, even those which are arbitrary members of these genders, I have argued that different flavors of a given gender are indistinguishable in syntax because their pairings with roots are regulated by semantic licensing conditions which apply at LF. This is a departure from Kramer (2015) where it is proposed that only combinations involving interpretable gender flavors are subject to LF licensing conditions.

I have presented evidence that only conjoined singulars trigger resolution on &P in Xhosa, because it is the last resort approach to providing agreement with conjuncts. Where possible Multiple Agree values the probe, but this works only if the number features of &P and the conjuncts match.

Though I have adopted much of a percolation and conversion approach to agreement resolution for conjoined singulars proposed in Adamson & Anagnostopoulou (2024), an important aspect of their system does not play a role in my account: feature hierarchies and the dependency relations within them, as discussed briefly in §5. Instead, I have argued that in the Bantu languages of this study, default values are arrived at by peeling away outer *nP*-layers that overlay the *nPs* of the semantic primitives.

Relatedly, the Xhosa and Shona patterns differ from those of the three-gender systems that Adamson & Anagnostopoulou analyze in the fact that [human] functions as a default. Adamson & Anagnostopoulou argue for the generalization in (136), which clearly does not hold in these Bantu languages.

- (136) Animacy Generalization: genders linked to non-animate reference are featurally less specific...morphosyntactic default environments can give rise to exponence that realizes the features of the non-animate gender, but not the features that realize the animate...(Adamson & Anagnostopoulou 2024 on defaults in 3-gender languages)

Under my approach in which Bantu's resolved agreement is not determined by working from most to least specific along a feature hierarchy as in the languages of their study, the irrelevance of (136) for Bantu requires no further explanation and need not be stipulated. The Animacy Generalization reflects an organizational pattern guiding *nP* stacking in the 3-gender

languages which in turn reflects well-motivated feature hierarchies; Bantu genders are organized along different lines.

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